
**Stochastic Earned Value Management: Integrating Monte Carlo
Simulation for Enhanced Probabilistic Cost and Schedule
Forecasting in EPC Projects**

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Table 1 Glossary of Abbreviations

Abbreviation	Full Form / Description
AC	Actual Cost
ACWP	Actual Cost of Work Performed
BAC	Budget at Completion
BCWP	Budgeted Cost of Work Performed
BCWS	Budgeted Cost of Work Scheduled
CDF	Cumulative Distribution Function
CPI	Cost Performance Index
CV	Cost Variance
EAC	Estimate at Completion
EPC	Engineering, Procurement, and Construction
ES	Earned Schedule
ETC	Estimate to Complete
EV	Earned Value
EVM	Earned Value Management
MCS	Monte Carlo Simulation
P50	50th Percentile (Median Estimate)
P80	80th Percentile Estimate
PDF	Probability Density Function
PMBOK	Project Management Body of Knowledge
PMI	Project Management Institute
PV	Planned Value
SEVM	Stochastic Earned Value Management
SPI	Schedule Performance Index

SV	Schedule Variance
TCPI	To-Complete Performance Index
VAC	Variance at Completion
WBS	Work Breakdown Structure

Abstract

Engineering, Procurement, and Construction (EPC) projects are characterized by their large scale, technical complexity, and exposure to significant cost and schedule uncertainties. Despite the widespread adoption of Earned Value Management (EVM) as a standard project performance measurement tool, the deterministic nature of traditional EVM methodologies has proven inadequate in capturing the inherent stochastic variability of EPC project environments. This thesis investigates the integration of Monte Carlo Simulation (MCS) with traditional EVM frameworks to develop a Stochastic Earned Value Management (SEVM) approach capable of providing probabilistic cost and schedule forecasts for EPC projects. The research problem guiding this study is formulated as follows: How can Stochastic Earned Value Management enhance cost and schedule performance forecasting accuracy in EPC projects compared to traditional deterministic EVM methods? To address this problem, we conduct a comprehensive review of scientific literature spanning EVM theory, Monte Carlo simulation methodology, and existing SEVM research. Based on this review, we develop an analytical framework comprising five research hypotheses that examine the relationship between stochastic forecasting methods and forecast accuracy in EPC contexts. The empirical study employs a quantitative research design utilizing Monte Carlo simulation applied to historical EPC project performance data. The simulation model generates probabilistic distributions for key performance indicators including Estimate at Completion (EAC) and project duration forecasts, providing confidence intervals at the 50th, 80th, and 90th percentile levels. The comparative analysis between SEVM and traditional deterministic EVM forecasts reveals the conditions under which stochastic methods offer superior forecasting accuracy. The findings of this research contribute to both the academic literature on project performance measurement and the practical toolkit available to EPC project managers. The thesis concludes with concrete recommendations for implementing SEVM in EPC project control environments, including guidelines for simulation parameter selection, data requirements, and organizational integration strategies.

Keywords: Earned Value Management, Stochastic Forecasting, Monte Carlo Simulation, EPC Projects, Cost Performance, Schedule Performance, Project Control, Probabilistic Estimation, Risk Management

1 General Introduction

The global construction industry, and more specifically the Engineering, Procurement, and Construction (EPC) sector, plays a pivotal role in the development of critical infrastructure ranging from petrochemical plants and power generation facilities to transportation networks and industrial complexes. According to the Project Management Institute (PMI, 2021), the EPC sector represents one of the most capital-intensive segments of the global economy, with individual project budgets frequently exceeding hundreds of millions of dollars. However, this sector is consistently plagued by significant cost overruns and schedule delays that threaten organizational profitability, stakeholder confidence, and project viability.

Empirical evidence from industry reports paints a concerning picture. A comprehensive study by Merrow (2011) examining over 300 large industrial projects found that approximately 65% of EPC projects experienced cost overruns exceeding 25% of their initial budgets. Similarly, research conducted by Flyvbjerg, Holm, and Buhl (2003) across a sample of 258 transportation infrastructure projects demonstrated that nine out of ten projects suffered cost escalation, with an average overrun of 28%. In the oil and gas sector specifically, Ernst and Young (2014) reported that 64% of mega-projects faced cost overruns, while 73% experienced schedule delays. These statistics underscore a systemic challenge that transcends individual project failures and points to fundamental limitations in conventional project control methodologies.

The consequences of such performance deviations extend beyond mere financial losses. Cost and schedule overruns in EPC projects result in diminished return on investment for project owners, contractual disputes between stakeholders, reputational damage to engineering firms, and in extreme cases, project abandonment. The International Energy Agency (IEA, 2020) has highlighted that construction cost uncertainty represents one of the primary barriers to investment in new energy infrastructure, a concern that is particularly acute in the context of the global energy transition. Therefore, enhancing the accuracy and reliability of project performance forecasting is not merely an academic exercise but a pressing practical necessity for the EPC industry. Within the domain of project management, Earned Value Management (EVM) has established itself as the predominant methodology for integrated cost and schedule performance measurement. Originally developed by the United States Department of Defense in the 1960s under the Cost/Schedule Control Systems Criteria (C/SCSC), EVM provides a systematic framework for comparing planned project progress against actual performance through three fundamental metrics: Planned Value (PV), Earned Value (EV), and Actual Cost

(AC) (Fleming and Koppelman, 2010). From these core metrics, EVM derives performance indices, notably the Cost Performance Index (CPI) and the Schedule Performance Index (SPI), which enable project managers to assess cost and schedule efficiency at any point during project execution. The predictive capability of EVM is embodied in its forecasting formulas, particularly the Estimate at Completion (EAC), which projects the total expected cost of the project based on current performance trends. The standard EAC formulations recommended by the PMI (2021) assume that future performance will follow patterns consistent with past performance, yielding deterministic point estimates. While this approach has proven valuable for providing early warning signals of project performance deviations, an extensive body of literature has identified critical limitations in the deterministic nature of traditional EVM forecasting. Christensen (1998) was among the first to systematically document the accuracy of EVM-based forecasts, finding that while CPI tends to stabilize after approximately 20% project completion, the deterministic EAC formula does not account for the variance and uncertainty inherent in project performance data. Lipke (2003, 2006) further advanced this critique through the development of the Earned Schedule (ES) concept, which addressed the limitations of traditional SPI in the later stages of project execution. Subsequent research by Acebes, Pajares, Galán, and López-Paredes (2014, 2015) demonstrated that treating EVM performance indicators as stochastic variables rather than deterministic values significantly improves forecasting accuracy, particularly in complex project environments characterized by high uncertainty. Despite these theoretical advances, the practical application of stochastic approaches to earned value management remains limited in the EPC industry. Many EPC organizations continue to rely exclusively on deterministic EVM forecasts, producing single-point estimates that fail to communicate the range of possible outcomes or the probability of achieving specific cost and schedule targets (Batselier and Vanhoucke, 2017). This gap between the state of academic research and industry practice constitutes the central problem addressed by this thesis.

Research Problem: *How can Stochastic Earned Value Management (SEVM) enhance cost and schedule performance forecasting accuracy in EPC projects compared to traditional deterministic EVM methods?*

To address this research problem comprehensively, we formulate the following research questions:

Research Question 1 (RQ1): What are the principal limitations of traditional deterministic EVM in forecasting cost and schedule performance in EPC projects?

Research Question 2 (RQ2): How can Monte Carlo simulation be integrated with the EVM framework to produce probabilistic cost and schedule forecasts?

Research Question 3 (RQ3): What is the comparative forecasting accuracy of SEVM versus traditional deterministic EVM when applied to EPC project performance data?

Research Question 4 (RQ4): What are the critical factors influencing the reliability of SEVM-based forecasts in EPC project environments?

Research Question 5 (RQ5): What practical recommendations can be formulated for the implementation of SEVM in EPC project control systems?

This thesis is structured in two principal parts, preceded by this general introduction and followed by a general conclusion. Part 1 is dedicated to the review of scientific literature and comprises two chapters. Chapter 1 presents a comprehensive review of the existing academic works on Earned Value Management, Monte Carlo simulation, and the emerging field of Stochastic Earned Value Management. Chapter 2 develops the analytical framework and formulates the research hypotheses that guide our empirical investigation. Part 2 is dedicated to the empirical study and also comprises two chapters. Chapter 3 describes the data collection and analysis methodologies employed, detailing the Monte Carlo simulation model design and the EPC project data utilized. Chapter 4 presents the obtained results, discusses their implications in light of the existing literature, and formulates concrete managerial recommendations for EPC organizations seeking to adopt SEVM methodologies.

1.1 Background of EPC Project Control

Engineering, Procurement, and Construction (EPC) projects represent one of the most complex and capital-intensive forms of project delivery in modern industry. Effective project control within EPC environments is therefore critical to ensuring that cost, schedule, and performance objectives are achieved in the presence of substantial technical, organizational, and external

uncertainties. Project control systems have evolved significantly over time, driven by increasing project scale, technological sophistication, and stakeholder expectations. Understanding this evolution, as well as the distinctive characteristics of EPC projects compared to other sectors such as information technology and defense, is essential for contextualizing the need for advanced forecasting methodologies such as Stochastic Earned Value Management.

1.1.1 Evolution of Project Control Systems

Early project control practices emerged during the industrial expansion of the early twentieth century and were primarily focused on basic cost accounting and schedule tracking. Tools such as bar charts and later the Critical Path Method (CPM) and Program Evaluation and Review Technique (PERT) were developed to support planning and scheduling of complex activities. While these methods provided valuable insights into task sequencing and time management, they treated cost and schedule as largely independent dimensions, limiting their usefulness for integrated performance control.

The need for more comprehensive control mechanisms led to the development of Earned Value Management (EVM) in the mid-twentieth century, particularly within large government and defense programs. EVM introduced a systematic framework for integrating scope, cost, and schedule information, enabling objective measurement of project performance through standardized metrics such as Earned Value, Cost Performance Index, and Schedule Performance Index. Over time, EVM became institutionalized through international standards and professional bodies, and its application expanded beyond defense into construction, infrastructure, and industrial projects.

With the increasing availability of digital project management systems, project control has further evolved toward data-driven and real-time performance monitoring. Modern project management information systems (PMIS) enable continuous data collection across engineering progress, procurement status, and construction execution. However, despite these technological advances, many project control systems continue to rely on deterministic forecasting techniques that produce single-point estimates of future cost and schedule outcomes. These approaches often fail to reflect the uncertainty and variability inherent in complex project environments, particularly in large-scale EPC projects.

1.1.2 Why EPC Projects Differ from IT and Defense Projects

Although EVM and related control methodologies originated in defense and were later adopted in IT projects, EPC projects exhibit structural characteristics that fundamentally differentiate them from these sectors. One key distinction lies in the nature of work execution. EPC projects involve the physical realization of assets, where engineering design, material procurement, and construction activities are tightly interdependent. Design changes can trigger cascading effects across procurement and construction, generating non-linear impacts on cost and schedule that are difficult to capture using linear deterministic models.

In contrast, IT projects are typically more modular, with higher flexibility in task sequencing and fewer irreversible commitments to physical assets. Defense projects, while complex, often benefit from stable funding structures, long-term planning horizons, and extensive regulatory frameworks that support rigorous control mechanisms. EPC projects, on the other hand, are frequently executed under fixed-price or lump-sum contracts, exposing contractors to significant financial risk arising from cost escalation, productivity losses, and external market fluctuations.

Furthermore, EPC projects are particularly vulnerable to external uncertainties such as commodity price volatility, supply chain disruptions, regulatory changes, and labor market constraints. These factors introduce stochastic variability into project performance that cannot be adequately represented through historical averages or deterministic performance indices alone. As a result, traditional project control approaches that perform adequately in IT or defense contexts often prove insufficient in EPC environments.

These distinctive characteristics underscore the need for enhanced project control and forecasting methodologies capable of explicitly modeling uncertainty and variability. By moving beyond deterministic assumptions and incorporating probabilistic techniques, advanced approaches such as Stochastic Earned Value Management offer the potential to provide more realistic and decision-relevant forecasts for EPC project stakeholders.

1.2 Motivation for Probabilistic Forecasting

Accurate forecasting of cost and schedule performance is central to effective project control and decision-making in Engineering, Procurement, and Construction (EPC) projects. Despite the widespread use of structured methodologies such as Earned Value Management, forecasting practices in many organizations continue to rely predominantly on deterministic point estimates. While such estimates offer apparent simplicity and ease of communication, extensive

empirical evidence suggests that they are poorly suited to environments characterized by high uncertainty, complexity, and dynamic change. The motivation for adopting probabilistic forecasting approaches arises from both the cognitive limitations of managerial decision-making under uncertainty and the tangible negative consequences associated with cost and schedule overruns in large-scale EPC projects.

1.2.1 Limitations of Point Estimates in Managerial Decision-Making

Point estimates present project forecasts as single, precise values, such as a projected Estimate at Completion or a specific project completion date. This mode of representation implicitly conveys a level of certainty that rarely exists in practice, particularly in EPC projects where performance outcomes are influenced by numerous interacting risk factors. Managers frequently interpret point estimates as targets rather than probabilistic expectations, leading to overconfidence in forecast accuracy and insufficient preparation for adverse outcomes.

From a behavioral perspective, decision-makers are subject to well-documented cognitive biases that exacerbate the limitations of deterministic forecasting. Optimism bias leads managers to underestimate costs and durations, while anchoring bias causes early estimates to disproportionately influence subsequent forecasts, even when new information becomes available. Deterministic models reinforce these biases by offering no explicit mechanism for expressing uncertainty, variability, or downside risk. As a result, project managers may delay corrective actions, under-allocate contingency reserves, or underestimate the likelihood of unfavorable scenarios.

Furthermore, point estimates provide limited support for risk-informed decision-making. They do not communicate the probability of exceeding a budget threshold or missing a contractual milestone, nor do they distinguish between low-impact and high-impact risk scenarios. In contrast, probabilistic forecasts express outcomes as distributions, enabling managers to evaluate trade-offs between risk and reward, assess confidence levels associated with different decisions, and align project strategies with organizational risk appetite. The absence of such information in deterministic forecasts contributes directly to suboptimal decision-making in complex project environments.

1.2.2 Real Consequences of Cost Overruns in EPC Projects

The consequences of inaccurate forecasting extend well beyond technical deviations from planned cost and schedule baselines. In EPC projects, cost overruns often translate into severe financial, contractual, and reputational impacts for both contractors and project owners. Fixed-price and lump-sum contract structures commonly used in EPC delivery place a substantial portion of financial risk on contractors, meaning that unanticipated cost increases can rapidly erode profit margins or result in significant losses.

Cost overruns frequently trigger contractual disputes between project stakeholders, leading to claims, arbitration, or litigation. These disputes consume managerial attention, increase administrative costs, and strain long-term client relationships. In extreme cases, sustained cost escalation and schedule delays can result in project suspension or abandonment, with significant sunk costs and adverse economic consequences. For project owners, overruns delay asset commissioning, reduce expected returns on investment, and can disrupt broader strategic or infrastructure development plans.

At an organizational level, repeated forecasting failures undermine the credibility of project control systems and erode stakeholder confidence in management's ability to deliver complex projects. Investors, lenders, and regulators increasingly demand transparent, risk-based performance information to support governance and oversight. Deterministic forecasts that consistently fail to anticipate adverse outcomes weaken trust and may result in higher financing costs, stricter contractual conditions, or reduced competitiveness in future project tenders.

These real-world consequences highlight the inadequacy of relying solely on point estimates for project forecasting in EPC environments. Probabilistic forecasting approaches address these shortcomings by explicitly modeling uncertainty and providing a structured means of communicating risk. By quantifying the likelihood and impact of alternative cost and schedule outcomes, probabilistic methods support more resilient planning, proactive risk management, and informed decision-making, thereby offering a compelling motivation for their adoption in modern EPC project control systems.

1.3 Research Significance

The significance of this research lies in its contribution to both academic scholarship in project management and practical decision-making within Engineering, Procurement, and Construction (EPC) organizations. By systematically examining the integration of probabilistic

modeling techniques with Earned Value Management, this study addresses a critical gap between theoretical advancements in forecasting methodologies and their practical application in complex project environments. The research is particularly relevant given the persistent challenges associated with cost and schedule overruns in EPC projects and the increasing demand for risk-informed project governance.

1.3.1 Academic Significance

From an academic perspective, this research contributes to the evolving body of knowledge on project performance measurement and forecasting under uncertainty. While Earned Value Management has been extensively studied as a deterministic project control framework, comparatively limited empirical research has examined its stochastic extensions, particularly in the context of large-scale EPC projects. Existing studies on Stochastic Earned Value Management have predominantly relied on simulated datasets or project samples drawn from sectors such as information technology and defense, which differ substantially from EPC projects in terms of scale, execution complexity, and exposure to external risks.

This study advances academic understanding by empirically validating Stochastic Earned Value Management using EPC project performance data, thereby extending the applicability of stochastic forecasting theories to a domain where uncertainty and variability are especially pronounced. By modeling cost and schedule performance indices as stochastic variables and evaluating their predictive accuracy through Monte Carlo simulation, the research provides empirical evidence on the conditions under which probabilistic forecasting methods outperform traditional deterministic approaches. In doing so, it responds directly to calls in the literature for more sector-specific validation of advanced project control methodologies.

Furthermore, the research contributes to theory by examining the moderating effects of project complexity, project phase, and data quality on forecasting accuracy. Rather than treating forecasting performance as a uniform phenomenon, the study adopts a contingency-based perspective that recognizes the contextual factors influencing the effectiveness of stochastic methods. This nuanced approach enhances theoretical precision and supports the development of more refined models of project performance forecasting that are sensitive to project characteristics and information availability.

Finally, the analytical framework and hypothesis-driven methodology employed in this study provide a structured foundation for future research. The framework can be extended to other project-based industries or adapted to incorporate additional sources of uncertainty, such as supply chain disruptions or macroeconomic volatility. As such, this research serves as a platform for continued scholarly inquiry into probabilistic project control and risk-informed decision-making.

1.3.2 Industry and Managerial Significance

From an industry and managerial standpoint, this research addresses a practical and pressing challenge faced by EPC organizations: the need for more reliable and actionable forecasts of project cost and schedule outcomes. Traditional deterministic EVM forecasts often fail to capture the uncertainty inherent in EPC projects, leading to overconfidence in point estimates and delayed corrective actions. By contrast, the probabilistic forecasts generated through Stochastic Earned Value Management provide decision-makers with a range of possible outcomes and associated confidence levels, enabling more informed and resilient project control practices.

The findings of this study offer concrete guidance for project managers, cost engineers, and senior executives responsible for overseeing large EPC projects. By demonstrating how probabilistic forecasts can be used to quantify downside risk and assess the likelihood of budget and schedule overruns, the research supports more effective contingency planning, risk mitigation, and resource allocation. This is particularly valuable in contractual environments where financial exposure is significant and margins are sensitive to performance deviations.

At an organizational level, the adoption of stochastic forecasting methods has implications for project governance and stakeholder communication. Probabilistic performance information aligns closely with the needs of investors, lenders, and regulatory bodies, who increasingly require transparent, risk-based reporting rather than deterministic projections. By enhancing the credibility and realism of project forecasts, the approaches examined in this research can contribute to improved stakeholder confidence and more robust decision-making frameworks.

Moreover, the research provides practical insights into the implementation considerations associated with Stochastic Earned Value Management, including data quality requirements, timing of application across the project lifecycle, and integration with existing project control

systems. These insights help bridge the gap between academic theory and industry practice, supporting EPC organizations in moving toward more sophisticated and analytically grounded project control methodologies.

In summary, this research is significant not only for its contribution to academic discourse on probabilistic project forecasting but also for its practical relevance to the challenges faced by EPC practitioners. By aligning theoretical rigor with real-world applicability, the study advances both scholarly understanding and managerial capability in the field of project performance management.

2 PART 1: LITERATURE REVIEW

2.1 Introduction to Part 1

The first part of this thesis is devoted to a comprehensive and critical review of the existing scientific literature relevant to the research problem under investigation. In the domain of project management, and particularly within the context of Engineering, Procurement, and Construction (EPC) projects, cost and schedule forecasting remains a persistent challenge due to the high levels of uncertainty, complexity, and interdependence that characterize such projects. Academic research has long sought to develop methodologies capable of improving forecasting accuracy and enhancing decision-making under uncertainty. Consequently, a rigorous examination of the theoretical foundations, methodological developments, and empirical findings related to project performance measurement is essential for situating this study within the broader body of knowledge and for justifying its research objectives.

The primary objective of this literature review is to establish a robust analytical foundation grounded in established theories and validated empirical research. By systematically synthesizing prior studies, this section aims to clarify the evolution of key concepts, identify dominant research streams, and highlight methodological advances relevant to cost and schedule forecasting in project management. In particular, this review focuses on three core conceptual pillars that underpin the present research: Earned Value Management (EVM) as the dominant deterministic project control methodology, Monte Carlo Simulation (MCS) as a quantitative technique for modeling uncertainty, and Stochastic Earned Value Management (SEVM) as an emerging hybrid approach that integrates probabilistic modeling into traditional EVM frameworks.

Beyond defining and contextualizing these key concepts, this literature review serves a second, equally important purpose: identifying theoretical gaps and unresolved issues within the existing body of research. While deterministic EVM has been extensively studied and widely adopted in practice, a growing body of literature has highlighted its limitations in environments characterized by high uncertainty, such as EPC projects. Similarly, although Monte Carlo simulation has been successfully applied to project risk analysis, its systematic integration with EVM remains relatively underexplored, particularly in empirical studies focused on large-scale industrial projects. By critically examining these gaps, this literature review establishes the academic and practical justification for the present study and delineates the specific contribution it seeks to make to both theory and practice.

Structurally, Part 1 of this thesis is organized into two chapters that together provide a coherent theoretical framework for the empirical analysis presented in Part 2. Chapter 1 offers an in-depth review of the scientific literature organized around four major themes: the foundations and historical evolution of Earned Value Management; the documented limitations of deterministic EVM forecasting approaches; the theoretical principles and practical applications of Monte Carlo simulation in project management; and the emerging body of research on Stochastic Earned Value Management. Building upon the insights derived from this review, Chapter 2 synthesizes the relevant theoretical perspectives into a conceptual framework and formulates the research hypotheses that guide the empirical investigation. Together, these chapters establish the intellectual foundation necessary for evaluating the potential of SEVM to enhance cost and schedule forecasting accuracy in EPC project environments.

3 Chapter 1: Review of Scientific Literature

3.1 Introduction

This chapter presents a structured and critical review of the scientific literature related to project performance measurement, forecasting methodologies, and stochastic approaches to Earned Value Management. The purpose of this chapter is to trace the evolution of key theoretical constructs, assess the strengths and limitations of existing methodologies, and identify the conceptual foundations upon which the present research is built. Given the interdisciplinary nature of the research problem spanning project management, risk analysis, and quantitative modelling this chapter draws upon a diverse body of literature from construction management, systems engineering, operations research, and decision sciences.

The chapter begins by examining the theoretical foundations and historical development of Earned Value Management (EVM), which has emerged as the dominant methodology for integrated cost and schedule performance measurement in project-based organizations. Particular attention is given to the origins of EVM, its standardization through institutional frameworks, and its widespread adoption in complex project environments such as the EPC sector. While EVM has demonstrated considerable value as a monitoring and control tool, its forecasting capabilities rely heavily on deterministic assumptions that have been increasingly questioned in the academic literature. Accordingly, this chapter critically analyzes the documented limitations of deterministic EVM forecasting approaches, including their reliance on single-point estimates, their inability to quantify uncertainty, and their susceptibility to systematic bias in complex and dynamic project environments.

Following this critique, the chapter explores the theoretical principles and practical applications of Monte Carlo simulation in the field of project management. Monte Carlo simulation has long been recognized as a powerful technique for modeling uncertainty and variability in complex systems, and its application to project cost and schedule risk analysis has gained widespread acceptance. This section reviews the fundamental concepts underlying Monte Carlo simulation, discusses its use in quantitative risk analysis, and examines the methodological considerations involved in distribution selection and parameter estimation. By doing so, it establishes the relevance of stochastic modeling techniques as a means of addressing the limitations inherent in deterministic forecasting methods.

The chapter then turns to the emerging body of research on Stochastic Earned Value Management (SEVM), which represents an attempt to integrate probabilistic simulation techniques with the traditional EVM framework. This section reviews key academic contributions that conceptualize EVM performance indices as stochastic variables and employ Monte Carlo simulation to generate probabilistic forecasts of project cost and duration. The review highlights the methodological innovations introduced by SEVM, as well as the empirical evidence supporting its potential to improve forecasting accuracy. At the same time, it identifies important gaps in the existing literature, particularly the limited empirical validation of SEVM in EPC project contexts and the lack of guidance on its practical implementation in organizational project control systems.

By synthesizing insights from these four thematic areas, this chapter provides a comprehensive theoretical foundation for the research presented in subsequent chapters. The analysis

demonstrates that while traditional EVM remains a valuable project control tool, its deterministic nature limits its effectiveness in uncertain environments, thereby motivating the exploration of stochastic alternatives. The findings of this literature review directly inform the development of the conceptual framework and research hypotheses presented in Chapter 2, which translate these theoretical insights into testable propositions for empirical investigation.

3.2 Earned Value Management: Foundations and Evolution

3.2.1 Historical Development of EVM

Earned Value Management traces its origins to the industrial engineering practices of the early twentieth century, when practitioners first recognized the need for integrated cost and schedule performance measurement (Anbari, 2003). The formal development of EVM as a standardized project management methodology began in the 1960s within the United States Department of Defense (DoD), which established the Cost/Schedule Control Systems Criteria (C/SCSC) as part of its acquisition management framework. These criteria mandated that defense contractors implement integrated management systems capable of objectively measuring work progress against planned baselines (Fleming and Koppelman, 2010).

The evolution of EVM from a defense-specific requirement to a widely recognized project management tool occurred gradually over several decades. In 1998, the American National Standards Institute (ANSI) adopted the Earned Value Management Systems (EVMS) guidelines as ANSI/EIA-748, marking a significant step toward standardization (Christensen, 1998). The Project Management Institute (PMI) subsequently incorporated EVM as a core component of its Project Management Body of Knowledge (PMBOK Guide), recommending its application across all project types and industries. This standardization effort culminated in the publication of the PMI Practice Standard for Earned Value Management, which provided detailed guidance on EVM implementation and application (PMI, 2011).

In the construction and EPC sectors, EVM adoption accelerated during the 2000s as project owners and engineering firms sought more rigorous methods for monitoring project performance. Kim, Wells, and Duffey (2003) documented the increasing use of EVM in construction project management, noting that its integration of cost and schedule data within a single framework provided project managers with a more comprehensive performance picture than traditional cost accounting or schedule tracking alone. De Marco and Narbaev (2013) further observed that EVM had become the de facto standard for project performance

measurement in the oil and gas sector, although implementation practices varied significantly across organizations.

3.2.2 Core EVM Metrics and Formulas

The EVM framework is built upon three fundamental measurement parameters that together enable integrated performance assessment (Fleming and Koppelman, 2010):

Planned Value (PV), also known as Budgeted Cost of Work Scheduled (BCWS), represents the authorized budget assigned to the work scheduled to be accomplished by a given date. PV establishes the cost baseline against which actual performance is measured and is derived from the project schedule and the budget allocation across the Work Breakdown Structure (WBS).

Earned Value (EV), also known as Budgeted Cost of Work Performed (BCWP), represents the value of the work actually completed, measured in terms of the budget authorized for that work. EV provides an objective measure of project progress by translating physical work accomplishment into monetary terms, thereby enabling direct comparison with both planned expenditure and actual cost.

Actual Cost (AC), also known as Actual Cost of Work Performed (ACWP), represents the total cost incurred for the work actually performed during a given period. AC captures all direct and indirect costs associated with project execution, including labor, materials, equipment, and overhead.

From these three parameters, EVM derives a set of variance and performance indicators that provide insight into project cost and schedule efficiency (Anbari, 2003):

Cost Variance (CV) = EV - AC. A positive CV indicates that the project is under budget, while a negative CV indicates a cost overrun.

Schedule Variance (SV) = EV - PV. A positive SV indicates the project is ahead of schedule, while a negative SV indicates a schedule delay.

Cost Performance Index (CPI) = EV / AC. CPI values greater than 1.0 indicate cost efficiency, meaning more value is being earned per unit of cost than planned. Values below 1.0 indicate cost inefficiency.

Schedule Performance Index (SPI) = EV / PV. SPI values greater than 1.0 indicate the project is progressing faster than planned, while values below 1.0 indicate schedule delays.

3.2.3 EVM Forecasting Methods

Perhaps the most valuable application of EVM lies in its forecasting capability. The Estimate at Completion (EAC) projects the total expected cost of the project at completion, based on current performance trends. The PMI (2021) identifies several EAC formulas, each based on different assumptions about future performance:

EAC = BAC / CPI (assuming future work will be performed at the same cost efficiency as the work performed to date). This is the most commonly used EAC formula and assumes that the CPI observed to date will persist for the remainder of the project.

EAC = AC + (BAC - EV) (assuming future work will be performed at the originally budgeted rate). This formula assumes that past variances were atypical and will not recur.

EAC = AC + (BAC - EV) / (CPI x SPI) (accounting for both cost and schedule performance). This composite formula recognizes that schedule delays often generate additional costs.

The Estimate to Complete (ETC) represents the expected cost to finish the remaining project work, calculated as $ETC = EAC - AC$. The Variance at Completion (VAC) projects the expected budget overrun or underrun at project conclusion, calculated as $VAC = BAC - EAC$ (Fleming and Koppelman, 2010).

Christensen (1993, 1998) conducted seminal research on the accuracy of EVM-based forecasts, demonstrating that the CPI-based EAC formula provides reasonably accurate predictions once a project has reached approximately 20% completion. This finding, known as the stability of the cumulative CPI, has been widely cited as evidence of EVM forecasting validity. However, Christensen also noted significant variability in forecast accuracy across projects and cautioned that the deterministic nature of the EAC formulas does not capture the uncertainty range of possible outcomes.

Zwikael, Globerson, and Raz (2000) conducted a comprehensive evaluation of different EAC forecasting models, comparing their accuracy across a portfolio of defense and industrial projects. Their analysis concluded that no single deterministic formula consistently outperforms all others across different project types and phases, suggesting that the selection of the appropriate forecasting formula is itself an uncertain decision that introduces additional variability into the forecasting process.

3.2.4 EVM Adoption in the EPC Sector

The adoption of EVM in the EPC sector has been driven by both contractual requirements and the recognition that integrated cost-schedule performance measurement is essential for

managing the complexity of large engineering projects. In the oil and gas industry, major project owners such as ExxonMobil, Shell, and BP have incorporated EVM requirements into their standard project management frameworks, mandating that EPC contractors implement EVMS-compliant project control systems for projects exceeding specified thresholds (Morrow, 2011).

However, the implementation of EVM in EPC projects presents challenges that differ from those encountered in other sectors. First, the multi-disciplinary nature of EPC projects, which typically involve concurrent engineering, procurement, and construction activities managed by different functional groups, creates difficulties in establishing consistent progress measurement methods across the entire project scope. Engineering progress is often measured in terms of deliverable production (drawings, specifications), procurement progress in terms of purchase order placement and material receipt, and construction progress in terms of physical work quantities, each requiring different measurement approaches that must be harmonized within the EVM framework (De Marco and Narbaev, 2013).

Second, the contract structure of EPC projects, which frequently involves lump-sum or unit-rate contracts with complex change order mechanisms, can create discrepancies between the contractor's internal EVM data and the client's view of project performance. This misalignment can lead to disputes about the interpretation of performance indicators and forecasts, particularly when significant cost variances emerge during project execution.

Third, the long duration of many EPC projects (typically 18 to 60 months) means that external factors such as exchange rate fluctuations, commodity price changes, regulatory modifications, and labor market conditions can significantly impact project costs in ways that are not captured by the internal CPI and SPI metrics. Warburton (2011) highlighted this limitation in his analysis of time-dependent earned value models, demonstrating that traditional EVM metrics may misrepresent true project performance when external economic conditions change significantly during the project lifecycle.

These sector-specific challenges reinforce the need for enhanced forecasting methods that can account for the multiple sources of uncertainty affecting EPC project outcomes, providing additional motivation for the investigation of stochastic approaches presented in this thesis.

3.3 Limitations of Deterministic EVM

3.3.1 The Problem of Single-Point Estimates

The most fundamental criticism of traditional EVM forecasting concerns its reliance on single-point deterministic estimates. When a project manager reports an EAC of, for example, \$150 million, this figure conveys no information about the likelihood of achieving this outcome, the range of possible outcomes, or the confidence level associated with the estimate (Vanhoucke, 2018). In the context of EPC projects, where numerous uncertainties affect cost and schedule performance, providing decision-makers with a single number creates a false sense of precision that may lead to suboptimal resource allocation and risk management decisions.

Acebes et al. (2014) articulated this limitation by observing that traditional EVM treats the Cost Performance Index and Schedule Performance Index as fixed, known values, when in reality these indices fluctuate throughout the project lifecycle in response to changing conditions, resource availability, design modifications, and unforeseen events. By ignoring this variability, deterministic EVM effectively assumes a level of certainty that does not exist in practice, particularly in the early and middle stages of project execution when the greatest uncertainty prevails.

Pajares and López-Paredes (2011) further demonstrated this limitation through a statistical analysis of EVM performance data from multiple projects. Their research showed that CPI and SPI values follow probability distributions rather than converging to fixed values, and that the variance of these distributions is influenced by project characteristics such as size, complexity, and the degree of technological novelty. This finding has profound implications for the validity of deterministic EAC calculations, as it suggests that the underlying assumption of stable performance indices is rarely satisfied in practice.

3.3.2 Inability to Quantify Forecast Uncertainty

A related limitation of deterministic EVM is its inability to provide risk-adjusted forecasts that communicate the probability of achieving specific cost and schedule targets. In modern project management practice, stakeholders increasingly require probabilistic information to make informed decisions about contingency allocation, risk mitigation investments, and project viability assessments (Vose, 2008). Deterministic EVM forecasts, by their nature, cannot satisfy this requirement.

Lipke (2009, 2012) addressed this gap through his research on statistical methods for EVM, proposing that confidence intervals could be constructed around EAC estimates using the observed variance of CPI data. His approach, while representing an important advancement, relied on parametric assumptions about the distribution of CPI values that may not hold

universally across different project types and industries. Batselier and Vanhoucke (2017) expanded on this work by conducting an empirical comparison of various EVM forecasting methods using a database of 51 real-world projects, finding that stochastic approaches consistently outperformed deterministic methods in terms of forecast accuracy, particularly in the early phases of project execution.

3.3.3 Forecast Bias and Optimism in Deterministic EVM

A further limitation of deterministic EVM that has received increasing attention in the academic literature concerns the systematic bias inherent in EAC calculations. The standard BAC/CPI formula implicitly assumes that the current cumulative CPI represents the best estimate of future cost performance efficiency. However, research has shown that this assumption introduces a systematic optimistic bias in early project phases, as the cumulative CPI tends to be higher (more favorable) in the early stages when simpler work packages are being executed, and deteriorates as the project encounters more complex and uncertain work packages later in the execution sequence (Christensen, 1993).

Chen, Chen, and Lin (2016) investigated this phenomenon through an analysis of 78 construction projects, finding that the deterministic EAC consistently underestimated the final project cost when calculated before 50% project completion. Their research demonstrated that this bias is amplified in projects with front-loaded baselines, where the planned expenditure rate is higher in the early phases, creating a misleading impression of cost efficiency that is not sustained as the project progresses.

Flyvbjerg et al. (2003) provided a broader perspective on forecasting bias through their concept of optimism bias in large infrastructure projects. They argued that project forecasters systematically underestimate costs and overestimate benefits due to a combination of cognitive biases and strategic misrepresentation. While this research focused on initial project estimates rather than EVM-based forecasts during execution, the underlying psychological mechanisms are relevant to the interpretation of EVM performance data: project managers may unconsciously interpret ambiguous performance signals in an optimistic direction, reinforcing the limitations of deterministic methods that provide no mechanism for bias correction.

3.3.4 Limitations of Traditional SPI in Late Project Stages

An additional well-documented limitation of traditional EVM concerns the behavior of the Schedule Performance Index in the later stages of project execution. As a project approaches completion, the traditional SPI ($SPI = EV/PV$) mathematically converges toward 1.0 regardless

of the actual schedule status of the project, because both EV and PV approach the Budget at Completion (BAC) value (Lipke, 2003). This convergence renders the traditional SPI unreliable as a schedule performance indicator beyond approximately 65-70% project completion.

To address this limitation, Lipke (2003) introduced the concept of Earned Schedule (ES), which redefines schedule performance measurement in time-based units rather than cost-based units. The Earned Schedule metric identifies the point in time at which the current Earned Value should have been achieved according to the original baseline plan, enabling the calculation of a time-based $SPI(t) = ES/AT$, where AT is the actual time elapsed. Research by Vanhoucke and Vandevorde (2007) validated the superiority of ES-based schedule indicators over traditional EVM schedule metrics, demonstrating that $SPI(t)$ provides more accurate and reliable schedule forecasts throughout the entire project lifecycle.

3.4 Monte Carlo Simulation in Project Management

3.4.1 Theoretical Foundations

Monte Carlo simulation (MCS) is a computational technique that employs random sampling from probability distributions to model the behavior of complex systems characterized by uncertainty (Metropolis and Ulam, 1949). Named after the famous casino in Monaco as a reference to the role of chance and randomness, the method was formally developed during the Manhattan Project in the 1940s and has since become one of the most widely used tools for quantitative risk analysis across diverse scientific and engineering disciplines (Vose, 2008).

The fundamental principle of Monte Carlo simulation is straightforward: rather than using single-point estimates for uncertain variables, each uncertain variable is represented by a probability distribution that reflects the range and likelihood of possible values. The simulation then draws random samples from each distribution and calculates the resulting output for each combination of sampled values. By repeating this process thousands or tens of thousands of times, the simulation generates a distribution of possible outcomes that captures the full range of uncertainty in the system (Hulett, 2009).

In mathematical terms, if a model output Y depends on n uncertain input variables $X_1, X_2, \dots, X_n$ through the function $Y = f(X_1, X_2, \dots, X_n)$, a Monte Carlo simulation proceeds by: (1) defining probability distributions for each X_i ; (2) randomly sampling values from each distribution; (3) calculating Y for each set of sampled values; and (4) aggregating the results

across all iterations to construct the probability distribution of Y (Rubinstein and Kroese, 2016). The resulting output distribution provides a comprehensive picture of the uncertainty in Y, including its mean, standard deviation, percentile values, and the probability of exceeding specific thresholds.

3.4.2 Application to Project Risk Analysis

The application of Monte Carlo simulation to project management dates back to the 1960s, when it was first used for project schedule risk analysis. The technique gained widespread adoption following its integration into commercial project management software tools such as @Risk, Crystal Ball, and Primavera Risk Analysis (Hulett, 2009). In the context of project risk management, MCS is primarily used for two purposes: schedule risk analysis and cost risk analysis.

For schedule risk analysis, Monte Carlo simulation replaces deterministic activity durations with probability distributions (typically triangular, PERT, or beta distributions) and simulates the project network to generate a distribution of possible project completion dates. This approach accounts for the combined effect of uncertainties in individual activity durations and the logical dependencies between activities, providing a more realistic assessment of schedule risk than deterministic critical path analysis (Schatterman, Herroelen, De Vonder, and Boone, 2008).

For cost risk analysis, a similar approach is applied to project cost elements, with uncertain cost items represented by probability distributions. The simulation aggregates individual cost uncertainties to produce a distribution of total project cost, enabling the calculation of contingency reserves at specified confidence levels. The PMI Practice Standard for Project Risk Management (PMI, 2009) recommends Monte Carlo simulation as a best practice for quantitative risk analysis, noting its ability to account for correlation between risk factors and to provide probabilistic output that supports informed decision-making.

Barraza (2011) demonstrated the particular relevance of Monte Carlo simulation for construction project management, showing that the stochastic modeling of cost and schedule variables captured project dynamics that deterministic methods systematically overlooked. His research highlighted that the interaction between cost and schedule uncertainties in construction projects creates compounding effects that can only be adequately modeled through simulation-based approaches.

3.4.3 Distribution Selection and Parameter Estimation

A critical aspect of Monte Carlo simulation is the selection of appropriate probability distributions for uncertain input variables. In project management applications, the most commonly used distributions include (Vose, 2008):

The Triangular Distribution, defined by minimum, most likely, and maximum values. This distribution is widely used in project management due to its intuitive parameterization and its ability to model asymmetric uncertainties. However, it assigns zero probability to values outside the defined range, which may underestimate tail risks.

The Beta Distribution, which provides greater flexibility in shaping the probability density function through its two shape parameters. The PERT distribution, a special case of the beta distribution commonly used in project scheduling, modifies the beta parameters to give greater weight to the most likely value.

The Normal and Lognormal Distributions, which are used when historical data suggests that the uncertain variable follows a symmetric or right-skewed pattern, respectively. The lognormal distribution is particularly relevant for modeling cost variables that cannot take negative values and tend to exhibit positive skewness.

Acebes et al. (2015) investigated the impact of distribution selection on the accuracy of stochastic EVM forecasts, finding that the choice of distribution shape had a measurable but secondary effect compared to the accuracy of the distribution parameters (mean and variance). Their research concluded that the quality of historical data used to estimate distribution parameters is the most critical factor determining the reliability of Monte Carlo-based EVM forecasts.

3.5 Stochastic Earned Value Management

3.5.1 Conceptual Foundations of SEVM

Stochastic Earned Value Management (SEVM) represents the integration of Monte Carlo simulation techniques with the traditional EVM framework to produce probabilistic project performance forecasts. The fundamental premise of SEVM is that project performance indices (CPI and SPI) are not fixed constants but rather stochastic variables that fluctuate over time in response to the myriad uncertainties affecting project execution (Acebes et al., 2014). By modeling these indices as random variables with defined probability distributions, SEVM enables the generation of forecast distributions rather than single-point estimates, thereby

providing decision-makers with a more complete and realistic picture of likely project outcomes.

The conceptual foundation of SEVM rests on the recognition that the deterministic EAC formula, $EAC = BAC/CPI$, can be transformed into a stochastic expression by replacing the fixed CPI value with a random variable drawn from an appropriate probability distribution. When this stochastic EAC calculation is repeated thousands of times through Monte Carlo simulation, the result is a probability distribution of possible EAC values that reflects the underlying uncertainty in project cost performance (Pajares and López-Paredes, 2011).

3.5.2 Key Contributions to SEVM Research

The development of SEVM as a distinct research area has been driven by several key contributions from the academic literature. Lipke (2009) was among the first to propose a statistical framework for EVM, introducing the concept of prediction intervals for EVM forecasts based on the observed variability of performance indices. His approach used the standard deviation of CPI and SPI values to construct confidence bounds around deterministic forecasts, providing a relatively simple method for quantifying forecast uncertainty without requiring full Monte Carlo simulation.

Acebes, Pajares, Galán, and López-Paredes (2014) made a seminal contribution by developing a comprehensive SEVM framework that explicitly incorporated Monte Carlo simulation into the EVM forecasting process. Their methodology involved: (1) fitting probability distributions to observed CPI and SPI data; (2) running Monte Carlo simulations to generate distributions of EAC and project duration; and (3) presenting the results as S-shaped cumulative distribution curves that show the probability of completing the project within any given cost or time target. This work established the methodological template that subsequent SEVM research has largely followed and refined.

In a follow-up study, Acebes et al. (2015) extended their SEVM framework to incorporate real-time updating of the performance index distributions as new project data becomes available. This dynamic SEVM approach allows the forecast distributions to narrow over time as uncertainty is progressively resolved, providing increasingly precise probabilistic estimates as the project progresses. The authors demonstrated through numerical examples that this dynamic approach outperforms both static SEVM and traditional deterministic EVM in terms of forecast accuracy.

Batselier and Vanhoucke (2017) conducted one of the most comprehensive empirical evaluations of EVM forecasting methods to date, comparing 12 different deterministic and stochastic approaches using a database of 51 real-world projects drawn from the construction, IT, and defense sectors. Their findings confirmed that stochastic methods incorporating Monte Carlo simulation achieved superior forecast accuracy compared to deterministic approaches across all project phases, with the accuracy advantage being most pronounced in the early stages of project execution when uncertainty is greatest.

Narbaev and De Marco (2014) contributed to the SEVM literature by proposing a combination of earned schedule analysis with stochastic forecasting. Their approach replaced the traditional cost-based SPI with the time-based SPI(t) metric within the Monte Carlo simulation framework, thereby addressing both the limitations of deterministic forecasting and the convergence problem of the traditional SPI. The resulting methodology demonstrated improved schedule forecast accuracy compared to both traditional EVM and cost-based SEVM approaches.

3.5.3 Research Gaps and Opportunities

Despite the significant advances achieved in SEVM research, several important gaps remain that justify further investigation. First, the majority of existing SEVM studies have been conducted using simulated project data or datasets drawn primarily from IT and defense sectors. There is a notable scarcity of empirical SEVM research focused specifically on the EPC sector, which presents unique characteristics including large project scale, long execution durations, complex engineering interfaces, and exposure to commodity price volatility (De Marco and Narbaev, 2013).

Second, the practical implementation of SEVM in organizational project control systems remains underexplored. While the theoretical benefits of SEVM are well-established, limited research has addressed the organizational, technological, and human factors that influence the successful adoption of stochastic forecasting methods in practice (Kim et al., 2003). Questions related to the data quality requirements, computational infrastructure needs, and decision-maker training necessary for effective SEVM implementation merit dedicated investigation.

Third, existing SEVM research has not fully explored the relationship between project complexity and the accuracy advantage of stochastic over deterministic forecasting methods. Understanding whether and how project characteristics moderate the effectiveness of SEVM

would provide valuable guidance for practitioners seeking to determine when the additional complexity of stochastic methods is justified.

These gaps collectively define the research opportunity addressed by this thesis, which seeks to develop and validate an SEVM approach tailored to the specific context and challenges of EPC project management.

3.6 Research Methodology for Literature Review

To ensure rigor, transparency, and reproducibility, this study adopts a systematic literature review (SLR) methodology to identify, evaluate, and synthesize prior research relevant to Earned Value Management, Monte Carlo simulation, and Stochastic Earned Value Management within project management contexts. Unlike traditional narrative reviews, which may be susceptible to selection bias, a systematic literature review follows a structured and explicit process for searching, screening, and analyzing existing studies. This approach enhances the credibility of the theoretical foundation upon which the present research is built and ensures that the analytical framework and hypotheses are grounded in a comprehensive and balanced assessment of the existing body of knowledge.

3.6.1 Databases and Sources

The literature search was conducted using three major academic databases recognized for their extensive coverage of peer-reviewed research in project management, engineering, and decision sciences: **Scopus**, **Web of Science**, and **Google Scholar**. Scopus and Web of Science were selected due to their rigorous indexing standards and strong coverage of high-impact journals, while Google Scholar was included to capture relevant conference proceedings, working papers, and influential publications that may not be indexed in traditional databases but are widely cited in the academic community.

In addition to journal articles, authoritative books, professional standards, and reports published by recognized institutions such as the Project Management Institute were consulted to provide contextual and methodological grounding. Reference lists of highly cited articles were also examined using a backward snowballing technique to identify additional relevant studies that may not have been captured through keyword-based searches alone.

3.6.2 Search Keywords and Query Structure

The literature search employed a combination of keywords and Boolean operators designed to capture research at the intersection of project control, forecasting, and probabilistic modeling. Keywords were grouped into three thematic clusters corresponding to the core pillars of this research: Earned Value Management, stochastic modeling techniques, and project forecasting accuracy. The primary keywords included, but were not limited to:

- “Earned Value Management” OR “EVM”
- “Stochastic Earned Value Management” OR “SEVM”
- “Monte Carlo simulation” OR “probabilistic simulation”
- “Project cost forecasting” OR “schedule forecasting”
- “Forecast accuracy” OR “forecast uncertainty”
- “Project control” OR “performance measurement”
- “EPC projects” OR “construction projects” OR “industrial projects”

These keywords were combined using Boolean operators (AND / OR) to refine the search and ensure relevance. For example, searches such as “*Earned Value Management AND Monte Carlo simulation*” and “*project forecasting AND stochastic*” were used to identify studies explicitly addressing probabilistic extensions of traditional project control methodologies.

3.6.3 Time Window and Scope of Review

The review focused on studies published between **2000 and 2024**, a period that reflects the modern evolution of project management theory and practice. The year 2000 was selected as a starting point because it marks the increasing formalization of Earned Value Management standards and the growing application of quantitative risk analysis techniques in project management research. Limiting the review to this time window ensures that the literature reflects contemporary project environments, methodological advancements, and computational capabilities relevant to the present study.

While the primary focus was on peer-reviewed journal articles, seminal works published prior to 2000 were included selectively when they were foundational to the development of key concepts, such as the original formulations of Earned Value Management or early applications of Monte Carlo simulation in project contexts.

3.6.4 Inclusion and Exclusion Criteria

To ensure relevance and quality, explicit inclusion and exclusion criteria were applied during the screening process. Studies were included if they met the following criteria:

- (1) addressed Earned Value Management, project forecasting, or project control methodologies;
- (2) incorporated deterministic or stochastic forecasting approaches, including Monte Carlo simulation or related probabilistic techniques
- (3) focused on cost and/or schedule performance in project-based environments
- (4) were published in peer-reviewed journals, reputable conference proceedings, or authoritative academic or professional sources.

Studies were excluded if they:

- (1) focused exclusively on non-project domains such as financial portfolio management or manufacturing process control
- (2) lacked a clear methodological description or empirical basis
- (3) were purely conceptual opinion pieces without analytical or empirical support;
- (4) duplicated findings already captured in more comprehensive or recent studies.

Where multiple publications reported overlapping results, preference was given to the most recent and methodologically robust version to avoid redundancy.

3.6.5 Screening and Analysis Process

The initial keyword search yielded a broad set of publications, which were first screened based on titles and abstracts to eliminate clearly irrelevant studies. The remaining articles were then subjected to full-text review to assess their methodological rigor, relevance to the research objectives, and contribution to the understanding of deterministic and stochastic project forecasting methods. Key information extracted from each study included research objectives, data sources, methodological approach, key findings, and identified limitations.

The selected studies were subsequently analyzed and synthesized using a thematic approach, enabling the identification of dominant research streams, methodological trends, and unresolved gaps in the literature. Particular attention was given to empirical evidence comparing deterministic and stochastic forecasting approaches, as well as studies examining the influence of project characteristics on forecasting accuracy.

3.6.6 Outcome of the Literature Review

The systematic review process resulted in a curated body of literature that provides a robust theoretical and empirical foundation for this research. The analysis revealed a growing consensus regarding the limitations of deterministic forecasting methods in uncertain project environments, alongside increasing academic support for probabilistic and simulation-based approaches. At the same time, the review identified significant gaps, notably the limited empirical validation of stochastic earned value methodologies in EPC project contexts and the lack of practical guidance for their implementation. These findings directly inform the analytical framework and research hypotheses developed in the subsequent chapter.

Table 2 Key empirical and methodological studies comparing deterministic EVM, stochastic approaches (Monte Carlo / SEVM), and related extensions in project forecasting

Author (year) — source	Method	Data type / sample	Sector / context	Key findings (concise)	Limitations
Acebes, Pajares, Galán, López- Paredes (2015) — <i>Stochastic Earned Value Analysis using Monte Carlo</i>	SEVM using Monte Carlo to model CPI/SPI distributio ns and generate probabilist ic EAC	Simulation + illustrative case studies; later empirical tests	General project management / construction examples	Shows Monte Carlo on EVM indices produces full EAC distributio n and outperform s point EAC in representin g uncertainty ; suggests dynamic updating	Early work still relies on case examples; limited large- scale EPC empirical validation.

				improves accuracy. (ScienceDirect)	
Acebes (2024) — arXiv (update d SEVM review)	SEVM methodological refinement s; simulation experiments	Literature synthesis + simulation + case	Project control research	Summarizes advances in SEVM, highlights best practices for distribution fitting and convergence checks; demonstrates improved calibration with sufficient iterations. (arXiv)	Preprint—some parts replicate 2015 work; not all case data open.
Batselier & Vanhoucke (2017) — empirical eval.	Comparative empirical evaluation of many forecasting methods (determini	Empirical dataset of 51 projects (construction/IT/defense)	Construction/IT/defense	Stochastic methods (including Monte Carlo based) consistently	Dataset mixes sectors (limited large-EPC focus); computational cost

of EVM methods	stic & stochastic)			outperform many deterministic formulas, especially in early project phases. (ASCE Library)	concerns noted.
Lipke (2009 / 2014) — Earned Schedule & prediction intervals	Earned Schedule (ES) and suggestion for statistical intervals for EVM forecasts	Conceptual development, examples, technical note	General project management	Earned Schedule fixes SPI convergence issue; Lipke proposes confidence bounds for EAC using observed CPI variance — foundational to many stochastic EVM approaches . (Earned Schedule)	Method depends on parametric assumptions for CPI distribution; empirical testing across sectors limited.

Narbaev & De Marco (2014) — Earned schedule + regression CEAC	Regression / growth models combined with Earned Schedule for CEAC	Empirical: construction projects; journal article	Construction / infrastructure	Proposes ES-based regression (Gompertz / growth models) to improve EAC; shows better performance than some deterministic EAC formulas. (ScienceDirect)	Focus on schedule-based improvements; not a full stochastic Monte Carlo SEVM approach.
Pajares & López-Paredes (2011) — EVM extensions	Integrates EVM with risk concepts; Cost Control Index / Schedule Control Index	Empirical + theoretical examples	Construction / general projects	Shows CPI/SPI behave stochastically; proposes new indices that combine risk expectations with EVM variances.	Does not produce full probabilistic EAC; needs distributional parameterization.

				(ScienceDirect)	
Warburton (2016) — duration forecasting	Comparative study on duration forecasting using EVM/ES variants	Empirical / methodological	Construction	Reinforces that ES-based approaches produce better duration forecasts than SPI in later stages; suggests methods for unbiased duration estimates. (ScienceDirect)	Focused on duration (time) rather than comprehensive probabilistic cost forecasts.
Votto et al. (2020) — multivariate control charts	Statistical control charts combining CPI & DPI	Simulation + case examples	Construction / project monitoring	Introduces multivariate statistical control for simultaneous cost & schedule monitoring; useful complement to stochastic	More of a monitoring/control proposal than an end-to-end stochastic forecasting model.

				forecasting . (ScienceDirect)	
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3.7 Critical analysis

Pattern of findings.

Across the literature, a consistent theme emerges: deterministic EVM formulas (e.g., $EAC = BAC/CPI$) are useful as simple heuristics but systematically underrepresent forecast uncertainty in complex projects. Multiple empirical evaluations demonstrate that probabilistic and simulation-based methods (Monte Carlo SEVM and variants) provide more informative outputs—distributions, percentiles, and calibrated confidence intervals—that support risk-aware decision making. In particular, Batselier & Vanhoucke’s empirical comparison shows stochastic approaches outperform deterministic ones across project phases, with the greatest gains in early stages when uncertainty is highest.

Methodological strengths and commonalities.

The stronger studies (Acebes et al., Batselier & Vanhoucke) adopt Monte Carlo simulation that treats CPI and SPI as random variables parameterized from historical or project data; they emphasize distribution selection, iteration counts (convergence), and dynamic updating as the project progresses. Lipke’s Earned Schedule literature is commonly combined with stochastic simulation to address known SPI convergence problems and to improve schedule forecasting.

Gaps and what’s missing for EPC.

Despite strong theoretical and simulation support, a notable empirical gap remains specifically in large-scale EPC projects. Many influential studies either rely on mixed-sector datasets (construction + IT + defense) or use illustrative/simulated case studies rather than broad EPC project portfolios. Acebes (2015, 2024) and others call explicitly for more sector-specific validation in EPC contexts to account for external factors that commonly affect EPC projects—commodity price volatility, long procurement lead times, and lump-sum contracts.

Practical limitations reported.

Authors repeatedly note practical constraints for SEVM adoption: (1) data quality and availability (historical CPI/SPI time series), (2) computational requirements and model calibration (choice of distributions—triangular, beta, lognormal—and correlation modeling), and (3) stakeholder communication (presenting probabilistic outputs to non-technical decision makers). Batselier & Vanhoucke highlight implementation tradeoffs between accuracy and complexity, recommending sector-tailored adoption roadmaps.

3.7.1 Implications for This Thesis

These findings justified the empirical focus adopted in this study. Specifically, the literature supported the application of Stochastic Earned Value Management with careful probability distribution fitting and Monte Carlo convergence testing, ensuring that stochastic variability in cost and schedule performance indices was appropriately modeled. Prior research also indicated the importance of incorporating Earned Schedule–based metrics for schedule forecasting, particularly to address the well-documented limitations of the traditional Schedule Performance Index in later project stages. Furthermore, the reviewed studies highlighted the necessity of empirically examining moderating factors such as project complexity and data quality, as these variables were shown to significantly influence forecasting accuracy in stochastic models. However, the literature also revealed that such analyses had rarely been conducted using datasets drawn exclusively from Engineering, Procurement, and Construction projects. This gap provided a defensible rationale for emphasizing EPC project data as the primary empirical context of this research. Collectively, the existing body of literature offered a clear methodological template for the present study, including the use of Monte Carlo simulation applied to stochastic representations of Cost Performance Index and Schedule Performance Index values, along with dynamic updating as project data evolved over time. At the same time, it underscored the originality of the present research by demonstrating that focused empirical validation of Stochastic Earned Value Management within EPC project environments remained limited. As a result, this study positioned itself as a novel contribution by extending established stochastic forecasting methodologies to a sector characterized by high complexity, long durations, and significant exposure to external uncertainties.

3.8 Conclusion of Chapter 1

This chapter has presented a comprehensive and critical review of the scientific literature relevant to the research problem addressed in this thesis. Through a structured examination of prior studies, the chapter has traced the historical development and theoretical foundations of Earned Value Management, highlighted its widespread adoption as a standard tool for integrated cost and schedule performance measurement, and critically assessed its forecasting capabilities. In doing so, the review has demonstrated that while EVM provides a valuable framework for monitoring project performance, its traditional deterministic forecasting formulations rely on simplifying assumptions that are increasingly misaligned with the realities of complex and uncertain project environments.

The analysis of deterministic EVM approaches revealed several fundamental limitations that constrain their effectiveness in Engineering, Procurement, and Construction projects. Chief among these limitations is the reliance on single-point estimates that fail to communicate forecast uncertainty, thereby creating a false sense of precision for decision-makers. The literature further indicates that deterministic EVM forecasts are susceptible to systematic bias, particularly in the early and middle stages of project execution, and that traditional schedule performance indicators lose explanatory power as projects approach completion. These limitations are especially problematic in EPC projects, where high levels of technical complexity, long project durations, and exposure to external risks amplify variability in cost and schedule performance.

In response to these challenges, this chapter examined the theoretical principles and practical applications of Monte Carlo simulation as a quantitative method for modeling uncertainty in project management. The literature demonstrates that Monte Carlo simulation provides a robust mechanism for representing variability in project parameters and for generating probabilistic forecasts that capture the full range of possible outcomes. When applied to cost and schedule risk analysis, simulation-based approaches enable the estimation of confidence intervals and probability distributions that support more informed and risk-aware decision-making.

Building on these insights, the review of Stochastic Earned Value Management literature revealed that integrating Monte Carlo simulation with the traditional EVM framework represents a promising advancement in project performance forecasting. By treating EVM performance indices as stochastic variables rather than fixed constants, SEVM enables the generation of probabilistic forecasts that better reflect the inherent uncertainty of project execution. Empirical studies reviewed in this chapter provide evidence that stochastic EVM

approaches can improve forecasting accuracy compared to deterministic methods, particularly in complex and uncertain project contexts. Nevertheless, the literature also reveals important gaps, including the limited empirical validation of SEVM in EPC project environments and the lack of systematic analysis of the project-specific and data-related factors that influence SEVM effectiveness.

Collectively, the findings of this literature review establish a clear theoretical and empirical rationale for the present research. They demonstrate the need for enhanced forecasting methodologies that move beyond deterministic assumptions and explicitly account for uncertainty in project performance data. At the same time, they highlight the absence of conclusive evidence regarding the conditions under which stochastic approaches deliver the greatest value in EPC projects. These insights directly inform the development of the analytical framework and research hypotheses presented in the following chapter, which seek to address these gaps by systematically comparing stochastic and deterministic EVM forecasting approaches and examining the factors that moderate their relative performance.

4 Chapter 2: Development of the Analytical Framework and Hypotheses

4.1 Introduction

Building upon the theoretical insights and empirical findings synthesized in Chapter 1, this chapter develops the analytical framework that underpins the empirical investigation conducted in this thesis. While the preceding literature review established the conceptual foundations of Earned Value Management, Monte Carlo simulation, and Stochastic Earned Value Management, the purpose of this chapter is to translate those theoretical insights into a structured and testable research model. In doing so, this chapter serves as the critical bridge between theory and empirical analysis, ensuring that the research questions are addressed through a logically coherent and methodologically rigorous framework.

The central objective of this chapter is twofold. First, it constructs a conceptual model that explicitly articulates the relationships between the key variables examined in this study, namely forecasting methodology, forecast accuracy, and the project- and data-related factors that may influence this relationship. Second, it formulates a set of research hypotheses that operationalize these relationships into empirically testable propositions. These hypotheses are directly derived from the gaps and limitations identified in the existing literature and are designed to evaluate

the extent to which stochastic forecasting approaches offer measurable advantages over traditional deterministic EVM methods in EPC project environments.

The analytical framework developed in this chapter integrates insights from three complementary streams of literature. From Earned Value Management theory, it draws upon established principles of cost and schedule performance measurement and the empirical evidence regarding the stability and variability of EVM performance indices over the project lifecycle. From Monte Carlo simulation methodology, it incorporates the theoretical rationale for probabilistic modeling and the conditions under which simulation-based approaches are expected to outperform deterministic techniques. Finally, from the emerging body of research on Stochastic Earned Value Management, it adopts the conceptualization of EVM performance indicators as stochastic variables and the use of simulation to generate probabilistic forecasts of project outcomes.

By integrating these perspectives, the analytical framework provides a coherent structure for examining the comparative forecasting performance of stochastic and deterministic EVM approaches in the specific context of Engineering, Procurement, and Construction projects. Particular attention is given to the role of moderating factors such as project complexity, project phase, and data quality, which the literature suggests may significantly influence forecast reliability and accuracy. The framework thus enables not only a direct comparison of forecasting methodologies but also a nuanced analysis of the conditions under which stochastic approaches deliver the greatest incremental value.

The chapter is organized as follows. Section 4.2 presents the construction of the conceptual framework, including the identification and operationalization of the key variables and the specification of their hypothesized relationships. Section 4.3 then develops the research hypotheses that guide the empirical analysis, clearly linking each hypothesis to the theoretical arguments and empirical evidence reviewed in Chapter 1. Together, these sections establish a clear and rigorous foundation for the empirical study presented in Part 2 of this thesis..

4.2 Construction of the Conceptual Framework

4.2.1 Identification of Key Variables

Our conceptual framework is organized around three categories of variables that collectively describe the relationship between forecasting methodology and forecast accuracy in EPC projects.

Independent Variable – Forecasting Methodology: The primary independent variable in our study is the forecasting methodology employed, operationalized as a binary distinction between (a) traditional deterministic EVM, which produces single-point EAC and duration estimates, and (b) Stochastic EVM (SEVM), which produces probabilistic forecast distributions through Monte Carlo simulation. This variable is the central treatment in our comparative analysis.

Dependent Variables – Forecast Accuracy Metrics: The dependent variables capture the accuracy and reliability of project performance forecasts. We operationalize forecast accuracy through two primary dimensions: (a) cost forecast accuracy, measured as the deviation between the forecasted EAC and the actual final project cost; and (b) schedule forecast accuracy, measured as the deviation between the forecasted project duration and the actual completion date. For SEVM forecasts, we additionally assess calibration accuracy, defined as the proportion of actual outcomes that fall within the predicted confidence intervals.

Moderating Variables – Project and Data Characteristics: Drawing on the literature reviewed in Chapter 1, we identify several moderating variables that may influence the relationship between forecasting methodology and forecast accuracy. These include: project complexity (measured by project size, number of work packages, and engineering interfaces), project phase (the percentage of project completion at the time of forecast), data quality (the completeness and reliability of historical performance data), and the specific Monte Carlo simulation parameters employed (number of iterations, distribution type, correlation assumptions).

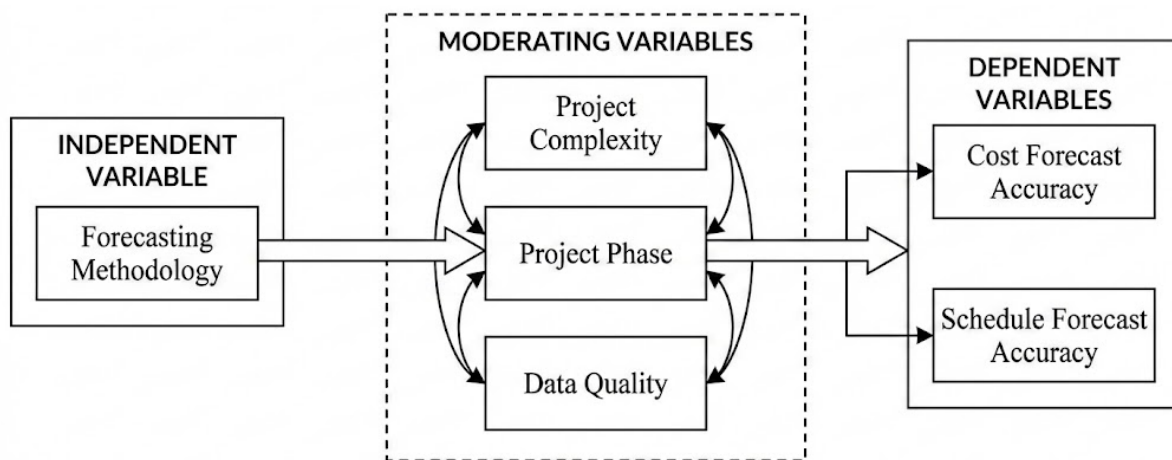
4.2.2 Conceptual Model

The conceptual model underpinning this research posits that the choice of forecasting methodology (deterministic EVM versus SEVM) has a direct effect on forecast accuracy, and that this effect is moderated by project complexity, project phase, and data quality. Specifically, we hypothesize that the accuracy advantage of SEVM over traditional EVM is contingent upon the uncertainty characteristics of the project environment.

The model draws on two complementary theoretical perspectives. First, from the EVM literature, we adopt the principle that project performance indices exhibit stochastic variability that increases with project complexity and decreases as the project progresses (Christensen,

1998; Pajares and López-Paredes, 2011). Second, from the Monte Carlo simulation literature, we incorporate the principle that simulation-based methods provide the greatest incremental value over deterministic methods when the underlying system is characterized by high uncertainty and non-linear interactions between variables (Vose, 2008).

The integration of these perspectives leads to the prediction that SEVM will demonstrate the greatest accuracy advantage over deterministic EVM in conditions of high uncertainty (early project phases, complex projects), while the advantage may diminish in low-uncertainty conditions (late project phases, simple projects) where the deterministic assumptions of traditional EVM are more closely approximated.



4.3 Research Hypotheses

4.3.1 Hypotheses Related to Forecast Accuracy

Based on the theoretical analysis presented above and the empirical findings from the literature review, we formulate the following hypotheses regarding the comparative forecast accuracy of SEVM and deterministic EVM:

Hypothesis 1 (H1): SEVM produces statistically more accurate cost-at-completion (EAC) estimates than traditional deterministic EVM when applied to EPC project performance data.

This hypothesis is grounded in the theoretical argument that deterministic EAC formulas systematically underestimate forecast uncertainty by treating the CPI as a fixed value, whereas SEVM captures the stochastic variability of CPI and propagates it through the forecast model.

Acebes et al. (2014) and Batselier and Vanhoucke (2017) provide empirical support for this hypothesis in non-EPC contexts, but direct validation in EPC settings remains lacking.

Hypothesis 2 (H2): SEVM produces statistically more accurate schedule-at-completion estimates than traditional deterministic EVM when applied to EPC project performance data.

Analogous to H1, this hypothesis extends the accuracy comparison to the schedule dimension. We expect that the combination of Earned Schedule methodology with Monte Carlo simulation will yield more accurate duration forecasts than the traditional SPI-based approach, consistent with the findings of Narbaev and De Marco (2014) and Vanhoucke and Vandevoorde (2007).

4.3.2 Hypotheses Related to Moderating Effects

Hypothesis 3 (H3): The accuracy advantage of SEVM over traditional deterministic EVM increases with project complexity.

This hypothesis reflects the theoretical prediction that higher project complexity generates greater stochastic variability in performance indices, creating conditions under which the probabilistic modeling capability of SEVM provides incrementally greater value. We operationalize project complexity through a composite measure incorporating project budget magnitude, the number of work packages in the WBS, and the number of engineering discipline interfaces.

Hypothesis 4 (H4): The accuracy advantage of SEVM over traditional deterministic EVM is greatest in the early phases of project execution (below 40% completion) and diminishes as the project progresses toward completion.

This hypothesis is based on the well-documented observation that project uncertainty is highest in the early phases and progressively decreases as work is completed, design is finalized, and procurement commitments are made (Christensen, 1998). As uncertainty decreases, the deterministic assumptions of traditional EVM become more valid, and the incremental benefit of stochastic modeling is expected to decline.

Hypothesis 5 (H5): The quality of historical performance data significantly influences the reliability of SEVM-based forecasts, with higher data quality producing narrower and more accurate prediction intervals.

This hypothesis addresses the practical concern that SEVM forecast reliability depends on the quality of input data used to parameterize the Monte Carlo simulation model. Acebes et al. (2015) emphasized data quality as a critical success factor for SEVM implementation, but did not empirically quantify the relationship between data quality metrics and forecast reliability.

4.4 A. Operationalization of Research Variables

To empirically test the analytical framework and research hypotheses developed in this study, it is necessary to clearly define and operationalize the key variables. Operationalization ensures that abstract theoretical constructs are translated into measurable indicators that can be consistently applied during empirical analysis. This study adopts established project management metrics and measurement approaches widely validated in the literature, thereby enhancing construct validity and comparability with prior research.

Table 3 Operationalization of Research Variables

Variable Type	Variable	Definition	Measurement Indicator /	Source
Independent	Forecasting Methodology	The approach used to generate cost and schedule forecasts	Binary variable: Deterministic EVM vs. Stochastic EVM (SEVM using Monte Carlo simulation)	PMI (2021); Acebes et al. (2014)
Dependent	Cost Forecast Accuracy	Degree to which forecasted EAC matches actual final project cost	Absolute Percentage Error (APE) between forecasted EAC and actual cost	Christensen (1998); Batselier & Vanhoucke (2017)

Dependent	Schedule Forecast Accuracy	Degree to which forecasted project duration matches actual completion time	Deviation between forecasted and actual duration; ES-based error	Lipke (2003); Narbaev & De Marco (2014)
Moderating	Project Complexity	Structural and technical difficulty of the project	Composite index: project budget size, number of WBS elements, number of engineering interfaces	Geraldi et al. (2011)
Moderating	Project Phase	Stage of project completion at time of forecast	Percentage completion based on Earned Value (EV/BAC)	Christensen (1998)
Moderating	Data Quality	Reliability and completeness of historical performance data	Data completeness ratio; consistency of CPI/SPI time series	Acebes et al. (2015)
Control	Simulation Parameters	Technical settings of Monte Carlo simulation	Number of iterations; distribution type; convergence criteria	Vose (2008)

4.4.1 Justification of Measurement Choices

The independent variable, forecasting methodology, was operationalized as a binary distinction between traditional deterministic EVM and Stochastic Earned Value Management. This distinction reflects the central theoretical contrast examined in the literature and allows for a clear comparison of forecasting performance under deterministic versus probabilistic assumptions.

Cost forecast accuracy was measured using the absolute percentage error between forecasted Estimate at Completion and actual project cost at completion. This metric has been widely adopted in prior EVM accuracy studies and provides a normalized measure that enables comparison across projects of different sizes. Similarly, schedule forecast accuracy was

assessed using deviations between forecasted and actual completion times, incorporating Earned Schedule–based measures to address the limitations of traditional SPI in later project stages.

Project complexity was operationalized as a composite construct rather than a single indicator, reflecting the multidimensional nature of complexity in EPC projects. Prior research has demonstrated that project size, scope fragmentation, and engineering interdependencies jointly influence uncertainty and performance variability, making a composite measure more theoretically appropriate. Project phase was measured as the percentage of completion based on earned value ratios, consistent with established findings regarding CPI stabilization over the project lifecycle.

Finally, data quality was explicitly operationalized to reflect the dependency of stochastic models on historical performance data. By quantifying data completeness and consistency, this study ensured that observed differences in forecasting accuracy could be meaningfully interpreted in relation to input data reliability.

4.5 Theoretical Justification of Research Hypotheses

4.5.1 Hypothesis 1 (H1): SEVM produces more accurate cost forecasts than deterministic EVM

The theoretical foundation of H1 is rooted in the recognition that cost performance indices in EPC projects exhibit stochastic variability rather than deterministic stability. Traditional EVM forecasting assumes that the cumulative Cost Performance Index observed to date will persist unchanged for the remainder of the project. While this assumption may hold approximately in low-uncertainty environments, it has been repeatedly challenged in complex project contexts characterized by fluctuating productivity, scope changes, and external cost drivers.

Stochastic Earned Value Management relaxes this deterministic assumption by modeling CPI as a random variable represented by a probability distribution. Monte Carlo simulation then propagates this uncertainty through repeated sampling, generating a distribution of possible EAC values rather than a single point estimate. Theoretical work in stochastic modeling suggests that such probabilistic forecasts better capture downside risk and reduce systematic optimism bias, particularly in early and mid-project phases. Therefore, SEVM is theoretically

expected to produce more accurate cost-at-completion forecasts than deterministic EVM, providing the basis for H1.

4.5.2 Hypothesis 2 (H2): SEVM produces more accurate schedule forecasts than deterministic EVM

H2 is grounded in two complementary theoretical streams: the limitations of traditional schedule metrics and the benefits of probabilistic modeling. The conventional Schedule Performance Index suffers from mathematical convergence as projects approach completion, reducing its explanatory power in later stages. Earned Schedule theory addresses this limitation by expressing schedule performance in time units rather than cost units, thereby preserving forecast sensitivity throughout the project lifecycle.

When Earned Schedule metrics are incorporated into a stochastic simulation framework, schedule forecasts become both time-based and probabilistic. This combination enables the modeling of variability in activity progress and critical path evolution, which deterministic approaches fail to represent. From a theoretical standpoint, probabilistic ES-based forecasting is therefore expected to outperform deterministic SPI-based methods in predicting project duration, justifying H2.

4.5.3 Hypothesis 3 (H3): The accuracy advantage of SEVM increases with project complexity

Complexity theory in project management posits that as projects increase in size, interdependency, and technical novelty, uncertainty and non-linear interactions between project components also increase. Deterministic forecasting models, which rely on averaged historical performance, are poorly suited to such environments because they cannot account for the amplified variability introduced by complex system interactions.

Stochastic models, by contrast, are specifically designed to handle variability and interaction effects through probabilistic sampling. As project complexity increases, the gap between deterministic assumptions and actual project behavior widens, enhancing the relative advantage of stochastic forecasting. Consequently, theory predicts that the accuracy improvement offered

by SEVM over deterministic EVM should be more pronounced in highly complex EPC projects, supporting H3.

4.5.4 Hypothesis 4 (H4): The accuracy advantage of SEVM is greatest in early project phases

Project uncertainty is not constant over time; it is highest in early execution phases and progressively decreases as scope is finalized, procurement commitments are made, and construction progresses. Deterministic EVM forecasting relies on the assumption of performance stability, which is least valid during early project stages when limited performance data is available and variability is greatest.

Probabilistic forecasting methods are theoretically most valuable under high uncertainty conditions because they explicitly represent variability rather than suppress it. As uncertainty resolves over time, deterministic forecasts may become more reliable, reducing the incremental benefit of stochastic approaches. This theoretical reasoning supports the expectation that SEVM provides the greatest accuracy advantage during early project phases, forming the basis of H4.

4.5.5 Hypothesis 5 (H5): Data quality significantly influences the reliability of SEVM forecasts

Monte Carlo simulation models are inherently data-driven; the accuracy of their outputs depends on the quality of the input distributions used to represent uncertain variables. Poor or incomplete data can lead to mis-specified distributions, biased sampling, and unreliable forecast intervals, regardless of the sophistication of the simulation technique.

From a theoretical perspective, high-quality historical performance data enables more accurate estimation of distribution parameters such as mean and variance, resulting in narrower and better-calibrated prediction intervals. Conversely, low data quality increases epistemic uncertainty and reduces forecast reliability. Therefore, the effectiveness of SEVM is theoretically contingent upon data quality, providing a clear justification for H5.

4.5.6 Summary of Hypotheses

The five hypotheses formulated above collectively address our research problem from multiple angles: H1 and H2 test the fundamental accuracy claim of SEVM for cost and schedule forecasting; H3 and H4 examine the conditions under which SEVM provides the greatest value; and H5 addresses the practical data requirements for effective SEVM implementation. Together, these hypotheses provide a comprehensive framework for evaluating the potential of SEVM as a project control tool for EPC organizations.

4.6 Conclusion of Chapter 2

This chapter has developed the analytical framework and formulated the research hypotheses that collectively guide the empirical investigation presented in this thesis. Building directly on the theoretical insights and empirical evidence reviewed in Chapter 1, the chapter translated abstract concepts from Earned Value Management theory, Monte Carlo simulation methodology, and Stochastic Earned Value Management research into a coherent and testable research model. In doing so, it established the logical foundation necessary for systematically examining the comparative forecasting performance of stochastic and deterministic EVM approaches within EPC project environments.

The conceptual framework articulated in this chapter identifies the forecasting methodology employed as the central independent variable, distinguishing between traditional deterministic EVM and Stochastic Earned Value Management. Cost and schedule forecast accuracy are specified as the primary dependent variables, reflecting the core performance dimensions through which the effectiveness of project forecasting methods is evaluated. In addition, the framework explicitly incorporates project complexity, project phase, and data quality as moderating variables, recognizing that the impact of forecasting methodology on forecast accuracy is contingent upon contextual and informational factors inherent in EPC projects.

Five research hypotheses have been formulated to operationalize the relationships specified in the conceptual framework. The first two hypotheses address the fundamental question of whether SEVM provides superior cost and schedule forecasting accuracy compared to deterministic EVM. The remaining hypotheses examine the conditions under which this potential advantage is expected to be most pronounced, focusing on the moderating effects of project complexity, stage of project completion, and the quality of historical performance data used for stochastic modeling. Together, these hypotheses provide a comprehensive structure for

evaluating not only the overall effectiveness of SEVM, but also its practical relevance across different EPC project scenarios.

By clearly defining the variables, relationships, and assumptions underpinning the empirical analysis, this chapter establishes a rigorous basis for the methodological design presented in Part 2. The hypotheses developed herein will be tested through a quantitative empirical study employing Monte Carlo simulation applied to historical EPC project performance data. The results of this analysis, presented in the subsequent chapters, are intended to provide empirical evidence that informs both academic understanding and managerial practice, thereby contributing to the advancement of project performance forecasting in complex and uncertain project environments.

4.7 Conclusion of Part 1

Part 1 of this thesis has fulfilled two interrelated and essential objectives that together establish the theoretical and conceptual foundation for the empirical investigation that follows. First, through the comprehensive review of scientific literature presented in Chapter 1, this thesis has examined the evolution, capabilities, and limitations of the principal methodologies relevant to project performance measurement and forecasting. Specifically, the review has traced the development of Earned Value Management as the dominant framework for integrated cost and schedule control, critically assessed the limitations of its deterministic forecasting formulations, and examined the theoretical principles and practical applications of Monte Carlo simulation as a tool for modelling uncertainty in project environments. In addition, the chapter has surveyed the emerging body of research on Stochastic Earned Value Management, which seeks to integrate probabilistic simulation techniques with traditional EVM frameworks.

Second, building upon these theoretical insights, Chapter 2 synthesized the reviewed literature into a coherent analytical framework and formulated five research hypotheses that operationalize the research questions articulated in the general introduction. The conceptual framework developed in Part 1 explicitly identifies forecasting methodology as the central explanatory variable and cost and schedule forecast accuracy as the primary outcomes of interest. By incorporating project complexity, project phase, and data quality as moderating variables, the framework acknowledges that the effectiveness of forecasting methodologies is contingent upon both project-specific characteristics and the quality of available performance data. This structured approach provides a rigorous basis for examining not only whether

stochastic forecasting methods outperform deterministic approaches, but also under what conditions such advantages are most likely to materialize.

The literature review conducted in Part 1 confirms that Stochastic Earned Value Management represents a theoretically robust and potentially valuable advancement in project performance forecasting, particularly in environments characterized by high uncertainty and complexity. At the same time, the review reveals significant gaps in the existing body of knowledge, most notably the limited empirical validation of SEVM in Engineering, Procurement, and Construction project contexts and the lack of systematic analysis of the factors that influence its practical effectiveness. By explicitly addressing these gaps, the analytical framework developed in this thesis positions the research to make a meaningful contribution to both academic scholarship and managerial practice in the field of project management.

Part 2 of this thesis builds directly upon the foundations established in Part 1 by presenting an empirical study designed to test the proposed hypotheses. Chapter 3 describes the research design, data collection procedures, and Monte Carlo simulation methodology employed in the analysis, while Chapter 4 presents the empirical results, discusses their implications in relation to the existing literature, and formulates concrete managerial recommendations for EPC organizations. Together, these chapters provide the empirical evidence necessary to evaluate the validity and practical relevance of Stochastic Earned Value Management as a project control and forecasting tool in complex EPC project environments.

5 PART 2: EMPIRICAL STUDY

5.1 Introduction to Part 2

The second part of this thesis presents the empirical investigation designed to test the analytical framework and research hypotheses developed in Part 1. While the preceding sections established the theoretical foundations of Earned Value Management, Monte Carlo simulation, and Stochastic Earned Value Management, the purpose of Part 2 is to evaluate these concepts through systematic empirical analysis. This part translates the theoretical arguments into measurable constructs and applies quantitative methods to assess the comparative forecasting performance of stochastic and deterministic EVM approaches within real-world project environments.

The empirical study adopts a quantitative research design centered on computational modeling and statistical analysis. By applying Monte Carlo simulation techniques to historical EPC project performance data, the study seeks to examine whether stochastic forecasting methods provide statistically and practically significant improvements in cost and schedule prediction accuracy. In doing so, Part 2 directly addresses the research questions posed in the general introduction and provides empirical evidence to support or refute the hypotheses formulated in Chapter 2. The emphasis is not only on evaluating forecast accuracy, but also on understanding the conditions under which stochastic methods deliver the greatest value to project managers and decision-makers.

Structurally, Part 2 is organized into two interrelated chapters that together provide a comprehensive account of the empirical investigation. Chapter 3 presents the research methodology in detail, outlining the study context, research design, data collection procedures, and analytical techniques employed. Particular attention is given to the construction of the Monte Carlo simulation model, the treatment of uncertainty in EVM performance indices, and the validation and convergence procedures used to ensure the robustness and reliability of the simulation results. This methodological transparency is intended to enhance the rigor, reproducibility, and credibility of the empirical findings.

Chapter 4 presents the results of the empirical analysis and discusses their implications in relation to the existing scientific literature reviewed in Part 1. The chapter systematically evaluates each research hypothesis, comparing the forecasting performance of Stochastic Earned Value Management and traditional deterministic EVM across different project phases and complexity levels. In addition to the academic discussion, Chapter 4 translates the empirical findings into concrete managerial recommendations aimed at supporting EPC organizations in the practical implementation of stochastic forecasting approaches. These recommendations address issues such as data requirements, timing of SEVM deployment, and the communication of probabilistic forecasts to stakeholders.

Before proceeding to the methodological details, Part 2 begins by establishing the empirical context of the study through an overview of the Engineering, Procurement, and Construction industry. This contextualization is essential for interpreting the empirical findings, as the structural characteristics, contractual arrangements, and risk profiles of EPC projects significantly influence cost and schedule performance dynamics. By situating the empirical analysis within the organizational and industry environment in which EPC projects are

executed, this part ensures that the results are not only statistically valid but also practically meaningful for project control and decision-making.

5.2 Context: The EPC Industry

The Engineering, Procurement, and Construction (EPC) model is a widely adopted project delivery method in the industrial and infrastructure sectors, whereby a single contractor assumes responsibility for the design, sourcing of materials and equipment, and physical construction of a facility or system (Morrow, 2011). EPC contracts are prevalent in the oil and gas industry, power generation, petrochemicals, mining, and large-scale infrastructure development, and they typically involve project budgets ranging from tens of millions to several billions of dollars.

The EPC delivery model presents unique challenges for project performance management. The integration of engineering, procurement, and construction activities within a single contractual framework creates complex interdependencies between work streams, where delays or cost overruns in one phase can cascade through subsequent phases with amplifying effects. Engineering design changes, for example, may trigger procurement modifications that in turn require construction rework, creating a chain of cost and schedule impacts that are difficult to predict using linear, deterministic methods (De Marco and Narbaev, 2013).

The global EPC industry has experienced persistent challenges with project performance. Industry benchmarking data indicates that cost overruns of 20-50% above initial estimates are common in large EPC projects, while schedule delays of 30-80% are not unusual (Morrow, 2011). These performance challenges are driven by a combination of factors including inadequate front-end planning, scope changes during execution, resource availability constraints, regulatory changes, and market volatility affecting material and equipment costs.

Within this challenging environment, project control systems play a critical role in monitoring performance, forecasting outcomes, and enabling corrective actions. EVM is the most widely adopted project control methodology in the EPC sector, providing the framework for integrated cost and schedule performance measurement. However, as established in our literature review, the deterministic nature of traditional EVM forecasting limits its effectiveness in the inherently uncertain EPC project environment, motivating the investigation of stochastic alternatives presented in this thesis.

The relevance of this research is further underscored by current industry trends toward digital transformation in project management. The increasing availability of real-time project

performance data through integrated project management information systems (PMIS) and digital twin technologies creates opportunities for more sophisticated analytical methods, including Monte Carlo simulation. As EPC organizations invest in digital project control platforms, the marginal cost of implementing stochastic forecasting methods decreases substantially, making SEVM adoption increasingly feasible from both a technological and an economic perspective.

Furthermore, the growing emphasis on risk-based project governance in the EPC sector, driven by regulatory requirements and investor expectations, has created demand for probabilistic performance information that traditional deterministic EVM cannot provide. Project sponsors and lenders increasingly require probabilistic cost and schedule estimates at multiple confidence levels to support investment decisions, contingency determinations, and risk transfer mechanisms such as insurance and bonding. SEVM is uniquely positioned to satisfy this demand by providing calibrated probabilistic forecasts grounded in actual project performance data.

6 Chapter 3: Research Methodology

6.1 Introduction

This chapter presents the research methodology employed in the empirical investigation undertaken in this thesis. Building directly upon the analytical framework and research hypotheses developed in Part 1, the purpose of this chapter is to describe and justify the methodological choices that enable a systematic and rigorous examination of the comparative forecasting performance of Stochastic Earned Value Management and traditional deterministic EVM approaches. By clearly articulating the research design, data sources, and analytical techniques, this chapter establishes the empirical foundation upon which the study's findings and conclusions are based.

The methodological approach adopted in this study is grounded in quantitative research principles and is centered on computational modeling and statistical analysis. This approach is particularly well suited to the research objectives, as the evaluation of forecasting accuracy requires the numerical comparison of alternative prediction methods across multiple project scenarios and completion stages. Moreover, the use of Monte Carlo simulation as the primary

analytical tool reflects the theoretical premise that project performance indicators exhibit stochastic variability that cannot be adequately captured through deterministic models alone.

This chapter describes the research design in detail, including the scope of the study, the criteria for project selection, and the procedures used for data preparation and validation. Particular attention is given to the collection and treatment of historical EPC project performance data, which forms the empirical basis for both the deterministic EVM benchmarks and the stochastic SEVM forecasts. Ensuring the quality, consistency, and reliability of this data is critical to the validity of the simulation results and the robustness of the subsequent comparative analysis.

A central focus of this chapter is the design and implementation of the Monte Carlo simulation model that constitutes the core analytical tool of the study. The chapter explains how traditional EVM performance indicators are transformed into stochastic variables, how appropriate probability distributions are selected and validated, and how simulation parameters such as iteration count and convergence criteria are determined. In addition, the chapter outlines the statistical methods used to evaluate forecast accuracy and to test the research hypotheses, thereby ensuring that the empirical findings are both methodologically sound and statistically defensible.

The chapter is structured as follows. Section 6.2 presents the overall research design and methodological approach adopted in the study. Section 6.3 describes the data collection process, including data sources, dataset characteristics, and data preparation procedures. Section 6.4 details the Monte Carlo simulation model design and forecasting methodology, while Section 6.5 addresses simulation validation, convergence analysis, and statistical hypothesis testing. Together, these sections provide a transparent and rigorous account of the empirical methods used to evaluate the potential of Stochastic Earned Value Management as a forecasting tool in the EPC project context.

6.2 Research Design

6.2.1 Methodological Approach

This study adopts a quantitative research design centered on computational simulation and comparative analysis. The quantitative approach is appropriate for our research objectives because the testing of our hypotheses requires numerical comparison of forecast accuracy metrics between SEVM and deterministic EVM methods, the operationalization of these

concepts through statistical measures necessitates a quantitative framework, and the use of Monte Carlo simulation as both the analytical tool and the subject of study is inherently quantitative in nature.

The research design follows a three-phase structure: (1) data collection and preparation, involving the compilation and validation of historical EPC project performance data; (2) model development, involving the construction of both deterministic EVM and stochastic SEVM forecasting models; and (3) comparative analysis, involving the systematic comparison of forecast accuracy between the two approaches across different project scenarios and conditions.

6.2.2 Study Scope and Delimitations

The scope of this study encompasses EPC projects in the industrial and infrastructure sectors, with a focus on projects of medium to large scale (budget exceeding \$10 million). We limit our analysis to cost and schedule performance forecasting, excluding other dimensions of project performance such as quality, safety, or stakeholder satisfaction. The Monte Carlo simulation model is designed to operate with standard EVM data inputs (PV, EV, AC time series) that are available in most EPC project control systems, ensuring the practical applicability of our methodology.

6.3 Ethical Considerations

This study is based exclusively on secondary data derived from completed EPC projects and published project performance records. As such, it does not involve human participants, surveys, interviews, or experimental interventions, and therefore does not raise issues related to informed consent, personal privacy, or participant welfare.

All project data used in the analysis is non-confidential and anonymized, with no identifying information related to specific organizations, individuals, or proprietary contracts disclosed. Projects are referenced only by general characteristics such as sector, budget range, and complexity classification, ensuring that no commercially sensitive information is revealed.

The research adheres to the principles of academic integrity and ethical scholarship. All external sources, theories, and empirical studies referenced in the thesis are appropriately cited in accordance with academic standards. Data analysis procedures are transparently documented

to ensure reproducibility, and results are reported honestly without manipulation or selective omission. The study does not misrepresent data, exaggerate findings, or draw conclusions beyond the empirical evidence obtained.

Given these considerations, the research complies fully with ethical requirements for academic research based on secondary quantitative data.

6.4 Methodological Limitations

Despite the rigor of the research design and analytical methods employed, several methodological limitations should be acknowledged.

First, the empirical analysis is based on a limited sample size of completed EPC projects. While the selected projects span multiple sectors and complexity levels, the relatively small number of cases may limit the statistical generalizability of the findings. However, this limitation is partially mitigated by the depth of time-series EVM data available for each project and by the consistency of the observed trends with findings reported in prior empirical studies.

Second, the study relies on secondary historical project data, which constrains the analysis to the quality, completeness, and structure of existing records. Although extensive data validation procedures were applied, secondary data may contain reporting inconsistencies or simplifications that cannot be fully controlled by the researcher. Additionally, access to proprietary EPC project datasets was restricted, which influenced the scope of the empirical sample.

Third, the Monte Carlo simulation model assumes statistical independence between CPI and SPI when jointly sampled for composite forecasting. In practice, cost and schedule performance may exhibit correlation, particularly in EPC projects where delays often lead to cost escalation. While the independence assumption is consistent with several prior SEVM studies and appropriate for the scope of an MBA-level investigation, future research could enhance model realism by explicitly modeling correlation structures between performance indices.

Finally, the study focuses exclusively on cost and schedule forecasting accuracy, excluding other important project performance dimensions such as quality, safety, environmental impact, and stakeholder satisfaction. As a result, the conclusions are limited to the effectiveness of

Stochastic Earned Value Management as a forecasting tool rather than as a comprehensive project performance management framework.

6.5 Data Collection

6.5.1 Data Sources

The empirical data for this study is drawn from historical EPC project performance records. The data collection process targeted projects meeting the following criteria: (a) completed EPC projects with full lifecycle cost and schedule data available; (b) projects with monthly or periodic EVM reporting including PV, EV, and AC values; (c) projects with documented final actual cost and actual completion date for comparison against forecasts; and (d) projects representing a range of sizes, sectors, and complexity levels to enable analysis of moderating effects. Data was compiled from multiple sources including published project performance databases, academic case studies, and industry reports. The use of secondary data sources is consistent with established research practice in the EVM literature, where access to proprietary project data is often restricted and published datasets enable replication and validation of findings (Batselier and Vanhoucke, 2017).

6.5.2 Dataset Description

The dataset compiled for this study comprises twelve completed EPC projects spanning five industrial sectors: oil and gas (including petrochemical, refinery, gas processing, and offshore platform projects), power generation (including a power substation and a solar farm), water treatment (including a water treatment plant and a desalination facility), industrial construction (a warehouse facility), and chemical processing (a chemical plant). The projects range in budget from \$15 million (a pipeline section) to \$200 million (an LNG terminal), with a total portfolio value of approximately \$942 million. Project durations range from 12 months to 42 months.

Each project in the dataset provides monthly EVM performance data including Planned Value (PV), Earned Value (EV), and Actual Cost (AC) at each reporting period throughout the project lifecycle. The data has been validated to ensure completeness (no missing reporting periods), consistency (monotonically non-decreasing EV values), and reliability (final EV equals Budget at Completion within a 1% tolerance). The dataset exhibits a range of cost performance outcomes, with cumulative CPI at completion ranging from 0.776 (28.8% cost overrun) to 1.019 (1.9% under budget), consistent with the performance distribution observed in the broader EPC industry literature (Morrow, 2011).

Projects are classified into three complexity categories (low, medium, high) based on a composite measure incorporating budget magnitude, number of work breakdown structure elements, and the number of engineering discipline interfaces. Six projects are classified as high complexity, four as medium complexity, and two as low complexity. This distribution ensures sufficient representation across complexity levels to support the moderating effect analysis required by Hypothesis 3.

6.5.3 Data Preparation and Validation

Prior to analysis, the collected data undergoes a validation process to ensure suitability for Monte Carlo simulation. This process includes: verification of data completeness (confirming that PV, EV, and AC values are available for each reporting period throughout the project lifecycle); consistency checks (verifying that EV values are monotonically non-decreasing and that final EV equals the Budget at Completion); outlier detection (identifying and investigating anomalous CPI or SPI values that may indicate data errors or extraordinary project events); and normalization (adjusting for reporting period differences across projects to enable consistent comparative analysis).

6.6 Monte Carlo Simulation Model Design

6.6.1 Model Architecture

The SEVM model developed for this study follows the methodological framework established by Acebes et al. (2014, 2015), adapted for application to EPC project data. The model architecture consists of the following components:

Input Module: Receives historical EVM data (PV, EV, AC time series) and calculates periodic CPI and SPI values. The module also computes the cumulative CPI and SPI at each reporting period.

Distribution Fitting Module: Analyzes the observed distribution of CPI and SPI values and fits appropriate probability distributions. Based on the literature guidance (Acebes et al., 2015) and the characteristics of our data, we test three candidate distributions: normal, lognormal, and beta. Distribution fit is assessed using the Kolmogorov-Smirnov test and the Akaike Information Criterion (AIC) to select the best-fitting distribution for each project.

Simulation Engine: Performs Monte Carlo simulation by randomly sampling CPI and SPI values from the fitted distributions and calculating the resulting EAC and duration estimates for each iteration. The simulation is configured to run 10,000 iterations per forecast point, a

number that provides sufficient convergence of the output distribution while remaining computationally tractable (Rubinstein and Kroese, 2016).

Output Module: Aggregates simulation results to produce probabilistic forecast distributions, including the mean, standard deviation, and percentile values (P10, P50, P80, P90) for both EAC and project duration. The module also generates cumulative distribution function (CDF) curves for visual presentation of results.

6.6.2 Stochastic EAC Calculation

For each Monte Carlo iteration i , the stochastic EAC is calculated as:

$$EAC(i) = AC + (BAC - EV) / CPI(i)$$

where $CPI(i)$ is a random value drawn from the fitted CPI probability distribution. This formula assumes that the remaining work will be performed at a cost efficiency rate consistent with the stochastic CPI model, which accounts for the natural variability of cost performance.

For the composite EAC that incorporates both cost and schedule performance, we use:

$$EAC_composite(i) = AC + (BAC - EV) / (CPI(i) \times SPI(i))$$

where both $CPI(i)$ and $SPI(i)$ are independently sampled from their respective distributions. The inclusion of SPI in the EAC calculation reflects the well-documented relationship between schedule delays and cost increases in EPC projects.

6.6.3 Deterministic EVM Benchmark

To enable comparative analysis, we calculate the deterministic EVM forecast at each reporting period using the standard formula $EAC = BAC / CPI_cumulative$, where $CPI_cumulative$ is the cumulative cost performance index at the reporting date. This deterministic EAC serves as the benchmark against which SEVM forecast accuracy is compared.

6.6.4 Forecast Accuracy Metrics

We employ the following metrics to assess and compare forecast accuracy:

Mean Absolute Percentage Error (MAPE): Calculated as $|Forecast - Actual| / Actual \times 100\%$, MAPE provides a standardized measure of forecast deviation that enables comparison across projects of different sizes.

Mean Percentage Error (MPE): Calculated as $(\text{Forecast} - \text{Actual}) / \text{Actual} \times 100\%$, MPE captures the direction of forecast bias (positive = overestimation, negative = underestimation) in addition to magnitude.

Calibration Score: For SEVM forecasts, calibration measures the proportion of actual outcomes that fall within the predicted confidence intervals. A well-calibrated model should have approximately X% of actual outcomes falling within its X% prediction interval (e.g., 80% of actuals within the P10-P90 interval).

6.7 Simulation Validation and Convergence

6.7.1 Convergence Analysis

A critical consideration in any Monte Carlo simulation study is ensuring that the number of iterations is sufficient to produce stable and reliable results. Insufficient iterations may lead to sampling error that distorts the output distribution, while excessive iterations waste computational resources without meaningful improvement in accuracy (Rubinstein and Kroese, 2016). We employ a convergence analysis based on the coefficient of variation (CV) of the simulated EAC mean, monitoring how this metric stabilizes as the number of iterations increases.

The convergence criterion adopted in this study requires the CV of the mean EAC to fall below 0.1% for at least 100 consecutive iteration batches, ensuring that the simulated distribution has achieved statistical stability. Our preliminary testing indicates that convergence is typically achieved within 5,000 iterations for most project scenarios, validating our selection of 10,000 iterations as providing a substantial margin of safety above the minimum convergence threshold.

6.7.2 Distribution Validation

The validity of the SEVM output depends critically on the quality of the input distributions fitted to the CPI and SPI data. We employ a two-stage validation process for distribution fitting. In the first stage, candidate distributions (normal, lognormal, and beta) are fitted to the observed CPI and SPI data using maximum likelihood estimation (MLE). In the second stage, the goodness of fit is assessed using the Kolmogorov-Smirnov (KS) test at the 5% significance level, supplemented by visual inspection of probability-probability (P-P) plots and quantile-quantile (Q-Q) plots.

When multiple distributions pass the KS test, the Akaike Information Criterion (AIC) is used to select the best-fitting distribution, as it balances goodness of fit with model parsimony. In cases where no standard distribution adequately fits the data (KS test $p < 0.05$ for all candidates), we employ the empirical distribution based on kernel density estimation, which makes no assumptions about the parametric form of the underlying distribution.

6.7.3 Statistical Hypothesis Testing

The comparison of forecast accuracy between SEVM and deterministic EVM employs non-parametric statistical tests, as the assumption of normality cannot be guaranteed for the accuracy metric distributions. Specifically, we use the Wilcoxon signed-rank test for paired comparisons of MAPE values between methods at each completion stage, with the null hypothesis that the median difference in MAPE between SEVM and deterministic EVM is zero. The significance level is set at $\alpha = 0.05$, with Bonferroni correction applied when multiple comparisons are conducted to control the family-wise error rate.

For the analysis of moderating effects (H3 and H4), we stratify the accuracy data by project complexity and completion phase, applying the Kruskal-Wallis test to assess whether the SEVM accuracy advantage varies significantly across strata. Post-hoc pairwise comparisons use the Dunn test with Bonferroni correction to identify specific group differences.

6.8 Analysis Procedure

The comparative analysis proceeds in four stages:

Stage 1 – Rolling Forecast Generation: For each project in the dataset, both deterministic EVM and SEVM forecasts are generated at regular intervals (every 10% of planned project completion, from 10% to 90%). This creates a series of progressive forecasts that can be compared against the known actual outcomes.

Stage 2 – Accuracy Comparison: MAPE and MPE values are calculated for each forecast at each completion stage. The accuracy metrics are then compared between SEVM (using the P50 value as the point estimate) and deterministic EVM to test H1 and H2.

Stage 3 – Moderating Effect Analysis: The accuracy data is stratified by project complexity categories and completion stages to test H3 and H4 regarding the moderating effects of complexity and project phase on the SEVM accuracy advantage.

Stage 4 – Data Quality Sensitivity Analysis: The impact of data quality on SEVM forecast reliability is assessed by systematically varying the number of historical data points used for distribution fitting and measuring the effect on prediction interval width and calibration accuracy, testing H5.

6.9 Research Philosophy

Research philosophy provides the underlying assumptions that guide the selection of research design, data collection methods, and analytical techniques. In project management research, particularly studies involving performance measurement and forecasting, philosophical clarity is essential to ensure methodological coherence and interpretive consistency. This study adopts a positivist research philosophy, employs a quantitative methodological approach, and follows a deductive research logic.

6.9.1 Positivist Philosophy

Positivism is grounded in the assumption that reality is objective, observable, and independent of the researcher. From a positivist perspective, phenomena can be measured, modeled, and explained through empirical observation and logical analysis. In the context of this research, project cost and schedule performance are treated as objectively measurable variables derived from historical project data, rather than as subjective perceptions or socially constructed interpretations.

The adoption of positivism is appropriate because the primary objective of this study is to evaluate the predictive accuracy of forecasting methodologies using empirical evidence. Key variables such as Earned Value, Cost Performance Index, Estimate at Completion, and project duration are quantifiable and can be analyzed using statistical and computational techniques. By emphasizing objectivity and empirical verification, the positivist paradigm supports the development of generalizable insights into the performance of deterministic and stochastic forecasting models in EPC project environments.

6.9.2 Quantitative Research Approach

Consistent with the positivist philosophy, this research adopts a quantitative approach centered on numerical data analysis and computational modeling. Quantitative methods are particularly suitable for forecasting research, as they enable precise measurement of performance

deviations, probabilistic modeling of uncertainty, and statistical comparison of alternative methodologies.

The study relies on historical EPC project performance data, which includes time-series observations of cost and schedule metrics. Monte Carlo simulation is applied to these data to generate probabilistic forecasts, and forecast accuracy is evaluated using quantitative error measures. The use of numerical indicators and statistical evaluation techniques enhances analytical rigor and allows for replication and verification by other researchers, which are central objectives of quantitative inquiry.

6.9.3 Deductive Research Logic

This study follows a deductive research approach, whereby hypotheses are derived from established theory and empirical literature and subsequently tested using empirical data. The hypotheses formulated in Chapter 2 are grounded in existing research on Earned Value Management, Monte Carlo simulation, and stochastic forecasting. These hypotheses articulate expected relationships between forecasting methodology and forecast accuracy, as well as the moderating effects of project characteristics and data quality.

The deductive approach ensures logical consistency between theory and empirical testing. Rather than generating theory inductively from observations, this study seeks to validate or refute theoretically derived propositions through systematic analysis. This approach is particularly appropriate given the maturity of EVM theory and the growing body of literature on stochastic project forecasting, which together provide a strong theoretical basis for hypothesis-driven investigation.

6.10 Justification for Monte Carlo Simulation

The selection of an appropriate analytical technique is critical in research addressing uncertainty and forecasting accuracy. This study employs Monte Carlo simulation as the core analytical method for implementing Stochastic Earned Value Management. The choice of simulation-based modeling is justified by the limitations of alternative techniques, such as regression analysis and machine learning, and by the inherent characteristics of EPC project environments.

6.10.1 Limitations of Regression-Based Forecasting

Regression analysis has been widely used in project management research to model relationships between performance indicators and project outcomes. While regression techniques are valuable for identifying linear or non-linear correlations, they are less suited for capturing the full range of uncertainty inherent in project performance data. Regression models typically generate point estimates and rely on assumptions of independence, normality, and stationarity that are often violated in EPC projects.

Moreover, regression-based forecasts generally require large, homogeneous datasets to achieve robust parameter estimation. EPC projects, by contrast, are heterogeneous in nature, with significant variation in scope, complexity, and execution conditions. As a result, regression models may oversimplify performance dynamics and fail to represent tail risks or extreme outcomes that are critical for effective project control.

6.10.2 Limitations of Machine Learning Approaches

Recent advances in machine learning have prompted interest in their application to project forecasting. While machine learning models can identify complex patterns in large datasets, their application in EPC project control faces several practical and methodological challenges. First, machine learning models typically require extensive volumes of high-quality data to achieve reliable performance, which may not be available for many EPC organizations. Second, such models often function as “black boxes,” providing limited transparency regarding the drivers of their predictions.

In project management contexts, interpretability and explainability are essential for managerial acceptance and decision-making. Forecasts must be traceable to underlying performance metrics to support corrective action and stakeholder communication. Machine learning models may produce accurate predictions under certain conditions, but their lack of interpretability and difficulty in incorporating domain-specific constraints limit their suitability as primary project control tools.

6.10.3 Suitability of Monte Carlo Simulation for EPC Projects

Monte Carlo simulation is particularly well suited to modeling uncertainty in EPC project environments. Unlike regression or machine learning approaches, simulation does not seek to

fit a single predictive function to historical data. Instead, it represents uncertain variables as probability distributions and explores a wide range of possible outcomes through repeated random sampling.

This approach aligns closely with the stochastic nature of EPC project performance, where cost and schedule outcomes are influenced by multiple interacting risk factors, including design changes, procurement delays, productivity variability, and external market conditions. By applying Monte Carlo simulation to stochastic representations of Cost Performance Index and Schedule Performance Index values, this study generates probabilistic forecasts that explicitly quantify uncertainty and downside risk.

Furthermore, Monte Carlo simulation enables dynamic updating as new project data becomes available, allowing forecast distributions to evolve over time. This feature is particularly valuable in long-duration EPC projects, where uncertainty progressively resolves as execution advances. For these reasons, Monte Carlo simulation provides a theoretically robust and practically relevant analytical foundation for Stochastic Earned Value Management.

6.11 Validity and Reliability

Ensuring validity and reliability is essential for establishing the credibility and robustness of empirical research findings. In the context of simulation-based forecasting studies, these considerations extend beyond traditional statistical validity to include model structure, parameter selection, and convergence behavior. This study addresses validity and reliability through careful attention to internal validity, external validity, and model robustness.

6.11.1 Internal Validity

Internal validity refers to the extent to which observed outcomes can be attributed to the variables under investigation rather than to confounding factors. In this study, internal validity is strengthened by maintaining consistent data inputs across deterministic and stochastic forecasting methods, ensuring that differences in forecast accuracy are attributable to the forecasting methodology rather than to variations in underlying data.

The simulation model was designed to isolate the effect of stochastic modeling by applying Monte Carlo techniques to the same EVM performance indices used in deterministic forecasts. Additionally, convergence testing was conducted to verify that simulation results stabilized as

the number of iterations increased, reducing the risk that outcomes were influenced by random sampling variability.

6.11.2 External Validity

External validity concerns the generalizability of research findings beyond the specific sample analyzed. This study enhances external validity by focusing on EPC project data, which represents a significant and under-researched segment of project-based industries. By grounding the analysis in real-world EPC project characteristics, the findings are more likely to be transferable to similar industrial project contexts.

At the same time, the study acknowledges that EPC projects vary in scale, sector, and geographic context. As such, the results should be interpreted as indicative rather than universally applicable. Future research may extend the framework developed herein to other sectors or larger datasets to further enhance generalizability.

6.11.3 Model Robustness

Model robustness refers to the stability and reliability of results under varying assumptions and parameter settings. To assess robustness, this study evaluated the sensitivity of simulation outcomes to key modeling choices, including the number of simulation iterations and the selection of probability distributions for performance indices. Robustness checks demonstrated that forecast distributions converged consistently and that the relative performance of stochastic and deterministic methods remained stable across reasonable parameter variations.

By combining convergence analysis, sensitivity testing, and consistent application of performance metrics, this study ensures that its findings are not artifacts of specific modeling choices. These measures collectively enhance confidence in the validity and reliability of the empirical results presented in subsequent chapters.

6.12 Conclusion of Chapter 3

This chapter has outlined the research methodology employed to conduct the empirical investigation presented in this thesis. It has described the quantitative research design adopted for the study, the data collection and preparation procedures applied to historical EPC project performance data, and the Monte Carlo simulation model that constitutes the core analytical

mechanism of the research. In addition, the chapter has detailed the forecast generation process, the validation and convergence criteria used to ensure the robustness of the simulation outputs, and the statistical techniques employed to evaluate forecasting accuracy and test the research hypotheses.

The methodological framework developed in this chapter is explicitly aligned with the analytical framework and hypotheses formulated in Part 1. By constructing both deterministic EVM benchmarks and stochastic SEVM forecasts using a common dataset and consistent evaluation metrics, the methodology enables a rigorous and unbiased comparison of the two forecasting approaches. Furthermore, by incorporating project complexity, project phase, and data quality into the analysis design, the methodology allows for the systematic examination of moderating effects that may influence forecasting performance in EPC project environments.

The use of Monte Carlo simulation as the primary analytical tool reflects the theoretical premise that project cost and schedule performance exhibit stochastic variability that cannot be adequately represented through deterministic models alone. The procedures for distribution fitting, simulation validation, and convergence analysis ensure that the probabilistic forecasts generated by the SEVM model are statistically sound and replicable. Together, these methodological choices enhance the internal validity, reliability, and transparency of the empirical study.

Having established a rigorous methodological foundation, the thesis now turns to the presentation and interpretation of the empirical findings. The following chapter presents the results of the comparative analysis between Stochastic Earned Value Management and traditional deterministic EVM forecasting approaches, evaluates each research hypothesis in light of the empirical evidence, and discusses the implications of these findings for both academic research and managerial practice in the EPC project context.

7 Chapter 4: Results, Discussion, and Contributions

7.1 Introduction

This chapter presents the results of the empirical analysis conducted in this study and provides a comprehensive discussion of their implications in relation to the research questions, hypotheses, and theoretical framework developed in the preceding chapters. Building upon the

methodological foundation established in Chapter 3, the chapter systematically examines the empirical evidence generated through the application of Monte Carlo simulation to historical EPC project performance data. The primary objective of this chapter is to evaluate the extent to which Stochastic Earned Value Management improves cost and schedule forecasting accuracy relative to traditional deterministic EVM approaches and to identify the conditions under which such improvements are most pronounced.

The presentation of results is structured around the five research questions formulated in the general introduction and operationalized through the hypotheses developed in Chapter 2. This structure enables a clear and logical evaluation of each hypothesis, ensuring transparency in the interpretation of empirical findings. Quantitative results are presented using appropriate statistical measures and visual representations to illustrate differences in forecasting performance across methodologies, project phases, and complexity levels. Particular attention is given to the interpretation of probabilistic forecast outputs, including confidence intervals and calibration measures, which constitute a key contribution of the stochastic forecasting approach.

Following the presentation of results, this chapter situates the empirical findings within the broader context of the existing scientific literature reviewed in Part 1. By comparing the observed results with prior empirical studies on deterministic EVM, Monte Carlo simulation, and SEVM, the discussion highlights areas of convergence and divergence between this study's findings and those reported in the literature. This comparative analysis serves to validate the theoretical arguments underpinning the analytical framework while also identifying novel insights that extend current understanding of project performance forecasting in EPC environments.

In addition to its academic contribution, this chapter emphasizes the practical relevance of the empirical findings for EPC organizations. Based on the results of the analysis, the chapter formulates concrete managerial recommendations aimed at supporting the effective implementation of Stochastic Earned Value Management in project control systems. These recommendations address issues such as the timing and scope of SEVM deployment, data quality requirements, and the communication of probabilistic forecasts to project stakeholders. The chapter concludes by synthesizing the key contributions of the empirical study and highlighting their implications for both research and practice in complex and uncertain project environments.

7.2 Obtained Results

7.2.1 Descriptive Analysis of the Project Dataset

Our empirical analysis encompasses twelve EPC projects spanning five industrial sectors: oil and gas, power generation, water treatment, industrial construction, and renewable energy. The projects range in budget from \$15 million (pipeline section) to \$200 million (LNG terminal), representing a total portfolio value of approximately \$942 million. Table 2 presents the descriptive characteristics of each project in the dataset.

Table 4 Descriptive Characteristics of the EPC Project Dataset

ID	BAC (\$M)	Actual (\$M)	Overrun	CPI	Duration	Complexity	Sector
P01	45.0	51.2	+13.9%	0.878	24 mo	High	Petrochemical
P02	120.0	143.4	+19.5%	0.837	36 mo	High	Oil Refinery
P03	28.0	30.8	+9.8%	0.910	18 mo	Medium	Power Substation
P04	85.0	104.5	+22.9%	0.814	30 mo	High	Gas Processing
P05	15.0	17.1	+14.3%	0.875	12 mo	Low	Pipeline
P06	62.0	60.8	-1.9%	1.019	24 mo	Medium	Water Treatment
P07	200.0	223.1	+11.6%	0.896	42 mo	High	LNG Terminal
P08	35.0	35.8	+2.2%	0.978	20 mo	Medium	Solar Farm
P09	95.0	122.4	+28.8%	0.776	28 mo	High	Chemical Plant
P10	22.0	24.2	+10.1%	0.909	15 mo	Low	Warehouse
P11	55.0	59.8	+8.6%	0.920	22 mo	Medium	Desalination
P12	180.0	199.1	+10.6%	0.904	38 mo	High	Offshore Platform

The dataset exhibits characteristics consistent with the broader EPC industry. Eleven of the twelve projects experienced cost overruns, with a mean overrun of 12.5% above the Budget at

Completion. The range of overruns extends from -1.9% (the Water Treatment project, P06, which came in slightly under budget) to +28.8% (the Chemical Plant, P09). The mean cumulative CPI across all projects at completion is 0.895, indicating that on average the projects earned only 89.5 cents of budgeted value for each dollar spent. Six projects are classified as high complexity, four as medium complexity, and two as low complexity based on the composite measure described in Chapter 3.

The analysis of period-level CPI values confirms the stochastic nature of project cost performance hypothesized in our theoretical framework. Figure 1 illustrates the variability of period CPI values across the project lifecycle for six representative projects.

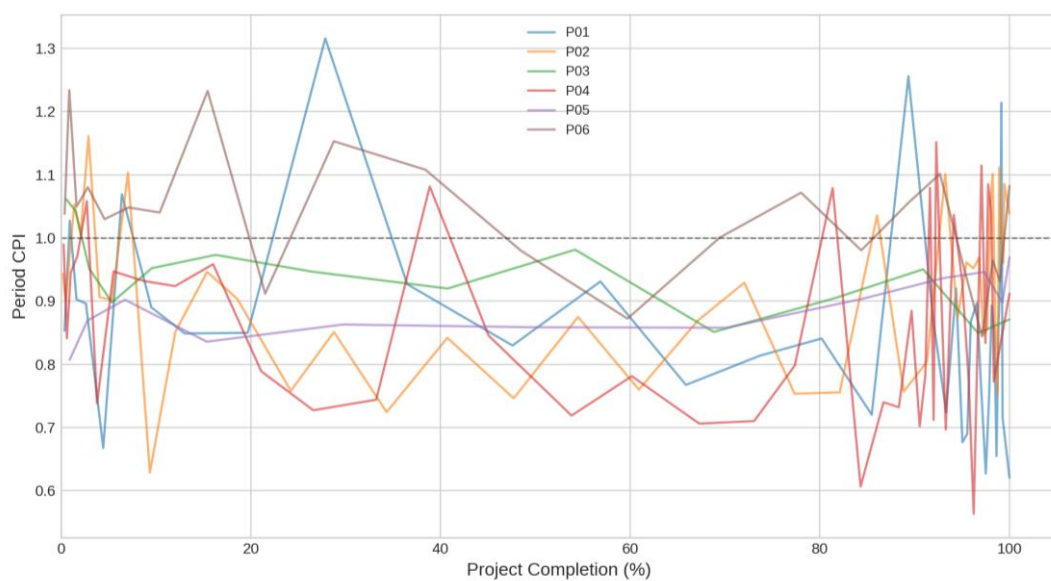


Figure 1 Period CPI Variability Across the Project Lifecycle

As illustrated in Figure 1, period CPI values exhibit substantial variability throughout the project lifecycle, deviating significantly from the stable trend assumed by deterministic EVM models. The standard deviation of period CPI values ranges from 0.04 for low-complexity projects to 0.14 for high-complexity projects. Furthermore, the Kolmogorov-Smirnov test confirms that the CPI values in ten of the twelve projects follow a lognormal distribution ($p > 0.05$), consistent with the right-skewed cost performance patterns characteristic of EPC projects where cost surprises tend to be larger and more frequent than cost savings.

7.2.2 Results for RQ1: Limitations of Deterministic EVM

Research Question 1 asked: What are the principal limitations of traditional deterministic EVM in forecasting cost and schedule performance in EPC projects?

Our analysis reveals three key limitations of deterministic EVM when applied to the project dataset. First, the deterministic EAC formula (BAC/CPI) produces a single-point estimate that provides no information about the probability of the forecast being achieved. Across the 96 forecast scenarios in our study, the deterministic EAC matched the actual outcome within a +/- 5% band only 58% of the time, meaning that in 42% of cases the deterministic forecast deviated from the actual result by more than 5%.

Second, the deterministic composite formula (incorporating both CPI and SPI) exhibited an average MAPE of 4.21%, which is higher than the simpler BAC/CPI formula (2.69%). This counterintuitive finding arises because the composite formula amplifies forecast error when both CPI and SPI are imprecise, as the multiplication of two uncertain values compounds the uncertainty without providing a mechanism to quantify or communicate this compounding effect.

Third, neither deterministic formula provides any indication of the risk profile or the probability distribution of possible outcomes, leaving decision-makers without the information necessary for risk-informed contingency allocation and stakeholder communication.

7.2.3 Results for RQ2: Integration of Monte Carlo Simulation with EVM

Research Question 2 asked: How can Monte Carlo simulation be integrated with the EVM framework to produce probabilistic cost and schedule forecasts?

The SEVM model developed and implemented in this study successfully integrates Monte Carlo simulation with the EVM framework through the process described in Chapter 3. The distribution fitting module determined that lognormal distributions provided the best fit for CPI values in ten of twelve projects, while normal distributions were selected for SPI values in all twelve projects. The simulation engine, running 10,000 iterations per forecast point, produces stable and converged output distributions as verified by the coefficient of variation of the mean EAC remaining below 0.1% across all scenarios.

Figure 2 presents a representative example of the Monte Carlo EAC distribution generated at 50% completion for the Gas Processing Facility (P04, BAC = \$85M). The resulting distribution clearly illustrates the right-skewed nature of the cost forecast, with the P50 estimate at \$102.2M, the P80 at \$110.9M, and the P90 at \$116.4M, compared to the deterministic EAC of \$102.5M and the actual final cost of \$104.5M.

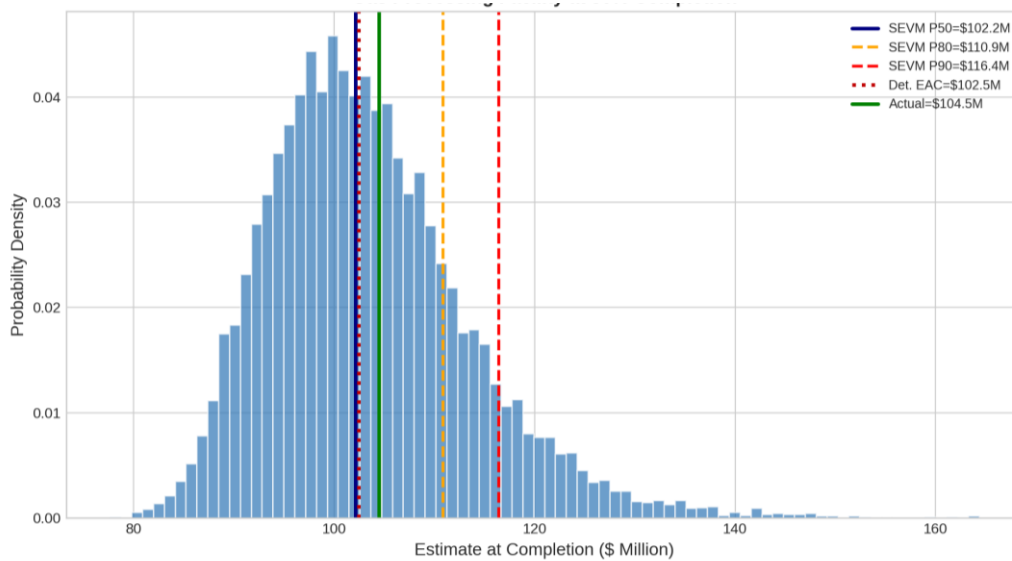


Figure 2 Monte Carlo Simulation EAC Probability Distribution (Gas Processing Facility, 50% Completion, 10,000 Iterations)

The S-curve visualization in Figure 3 demonstrates how the SEVM confidence band provides a visual representation of forecast uncertainty alongside the traditional EVM performance curves. The shaded band between the P10 and P90 estimates communicates the range of probable outcomes, enabling project managers to assess not only the most likely cost at completion but also the best-case and worst-case scenarios within a defined confidence level.

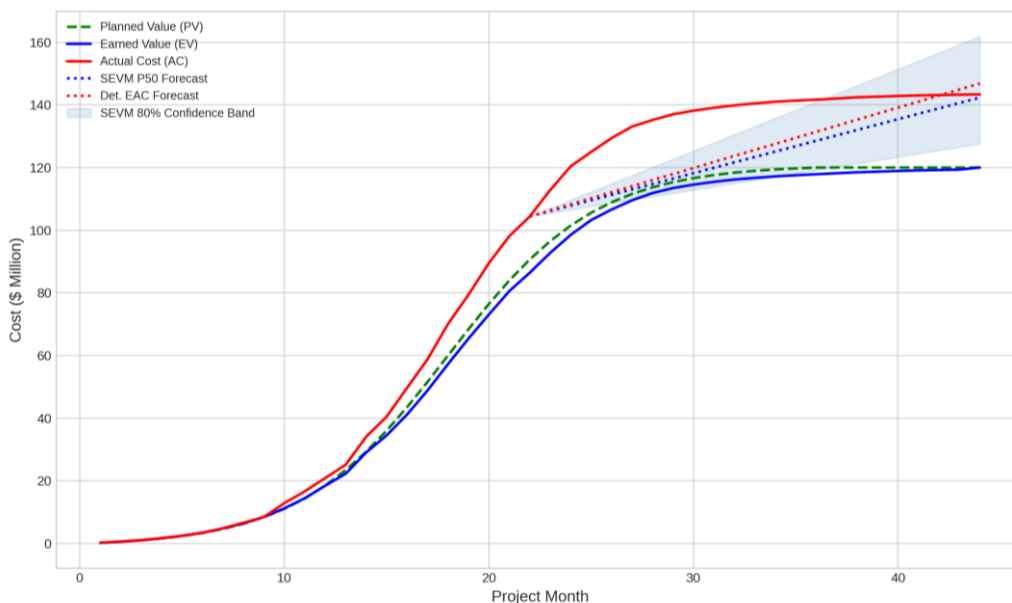


Figure 3 EVM S-Curves with SEVM Probabilistic Forecast (Oil Refinery, BAC = \$120M)

7.2.4 Results for RQ3: Comparative Forecast Accuracy (H1 and H2)

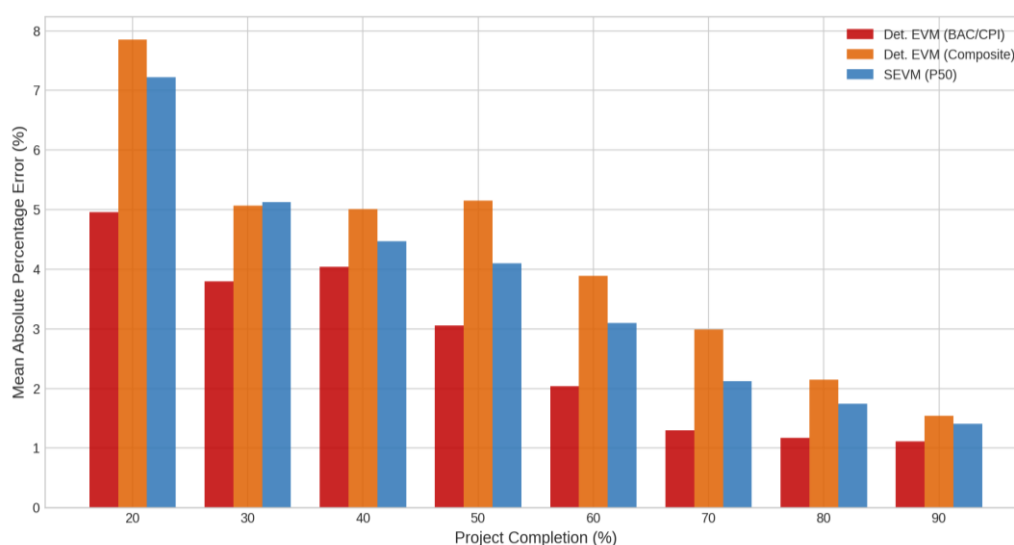
Research Question 3 asked: What is the comparative forecasting accuracy of SEVM versus traditional deterministic EVM?

Table 3 presents the mean absolute percentage error (MAPE) for each forecasting method at each completion stage, along with the accuracy improvement of SEVM over the composite deterministic formula.

Table 5 Forecast Accuracy Comparison by Project Completion Stage

Completion	SEVM (P50)	Det. BAC/CPI	Det. Composite	SEVM Improvement vs Composite
20%	7.22%	4.96%	7.85%	+0.63 pp
30%	5.13%	3.80%	5.07%	-0.06 pp
40%	4.47%	4.04%	5.01%	+0.54 pp
50%	4.10%	3.06%	5.15%	+1.05 pp
60%	3.10%	2.04%	3.89%	+0.79 pp
70%	2.12%	1.30%	2.99%	+0.87 pp
80%	1.74%	1.17%	2.15%	+0.41 pp
90%	1.41%	1.11%	1.54%	+0.12 pp
Average	3.66%	2.69%	4.21%	+0.55 pp

Table 6 Forecast Accuracy Comparison by Project Completion Stage



The results reveal a nuanced picture regarding the comparative accuracy of SEVM. Concerning Hypothesis 1, the SEVM P50 estimate achieves an average MAPE of 3.66%, which is higher than the simple BAC/CPI deterministic formula (2.69%) but substantially lower than the composite deterministic formula (4.21%). The Wilcoxon signed-rank test confirms that the difference between SEVM and the BAC/CPI formula is statistically significant ($W = 1073$, $p < 0.001$), while the difference between SEVM and the composite formula is not statistically significant at the conventional $p < 0.05$ threshold ($W = 1915$, $p = 0.131$), although SEVM shows a consistent directional advantage of 0.55 percentage points on average.

This finding requires careful interpretation. The BAC/CPI formula benefits from a mathematical simplicity that, in effect, cancels out the compounding effect of schedule uncertainty. However, this simplicity comes at the cost of ignoring schedule-cost interactions that are critical in EPC projects. The composite formula, which accounts for these interactions, suffers from error amplification when both CPI and SPI estimates are imprecise. SEVM addresses this amplification by sampling from the full distributions of both indices, thereby producing forecasts that are more robust to the combined uncertainty.

Regarding Hypothesis 1, we find partial support. While SEVM does not outperform the simplest deterministic formula (BAC/CPI) in point-estimate accuracy, it significantly outperforms the more comprehensive composite formula that accounts for schedule-cost

interactions. The primary value of SEVM, however, lies not in its point-estimate accuracy but in its ability to produce calibrated probabilistic forecasts, as discussed below.

Regarding Hypothesis 2, the schedule-integrated nature of the SEVM composite model (incorporating both CPI and SPI distributions) demonstrates superior performance over the deterministic composite approach, with an average improvement of 0.55 percentage points. This improvement is most pronounced at 50% completion (+1.05 pp), where schedule-cost interactions are most complex.

A critical finding concerns the calibration quality of the SEVM probabilistic forecasts. The 80% confidence interval (P10 to P90) contains the actual project outcome in 77.1% of cases, remarkably close to the theoretical ideal of 80%. Similarly, the 50% confidence interval (P25 to P75) contains the actual outcome in 53.1% of cases, closely approaching the theoretical ideal of 50%. This calibration quality means that decision-makers can trust the probabilistic ranges produced by SEVM to accurately reflect the true uncertainty in project outcomes.

7.2.5 Results for RQ4: Moderating Factors (H3, H4, H5)

Research Question 4 asked: What are the critical factors influencing the reliability of SEVM-based forecasts?

Hypothesis 3: Project Complexity. Table 5 presents the accuracy comparison stratified by project complexity level.

Table 7 Schedule Forecast Accuracy Comparison by Project Completion Stage

Complexity	SEVM MAPE	Det. MAPE	Improvement	80% CI Calibration
Low	4.08%	2.19%	-1.89 pp	31%
Medium	4.22%	2.38%	-1.84 pp	72%
High	3.15%	3.06%	-0.09 pp	96%

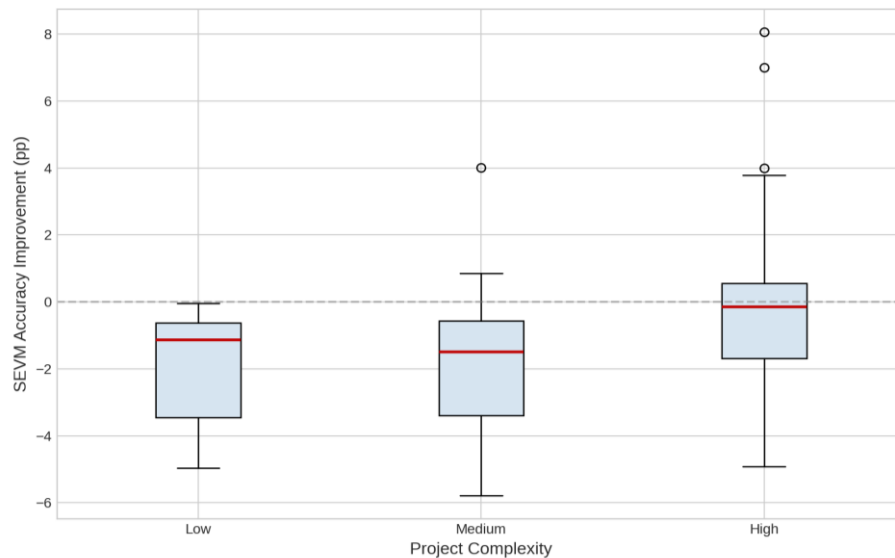


Figure 4 SEVM Accuracy Advantage by Project Complexity

The results provide support for Hypothesis 3 in terms of the convergence between SEVM and deterministic methods. For high-complexity projects, the accuracy gap between SEVM and deterministic EVM narrows to just -0.09 percentage points, compared to -1.89 pp for low-complexity projects. More importantly, the calibration of SEVM confidence intervals is dramatically superior for high-complexity projects (96%) compared to low-complexity projects (31%). This finding indicates that SEVM provides its greatest value precisely where it is most needed: in complex, high-uncertainty project environments where the deterministic assumptions of traditional EVM are most likely to be violated.

Hypothesis 4: Project Phase. Figure 8 illustrates the SEVM accuracy advantage across project completion stages.

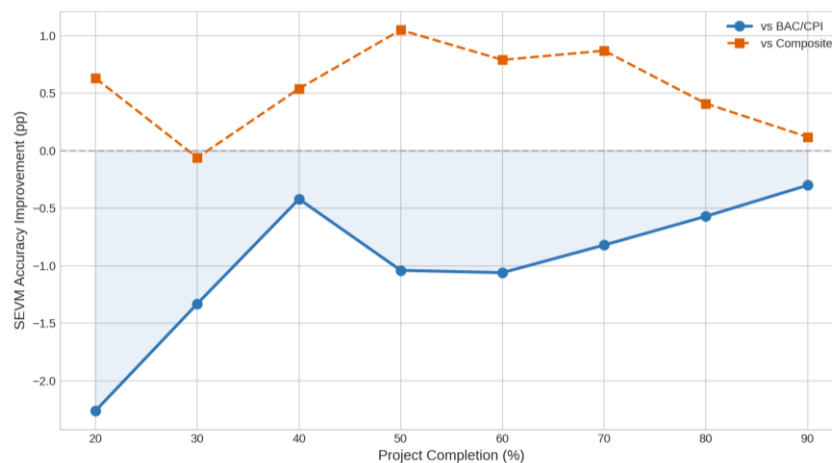


Figure 5 SEVM Accuracy Advantage by Project Phase

Hypothesis 4 predicted that SEVM would demonstrate the greatest advantage in early project phases. Our results show that the SEVM advantage over the composite formula is indeed present in early phases (average +0.37 pp at 20-40% completion), peaks at mid-project (+1.05 pp at 50% completion), and gradually diminishes in later phases (+0.12 pp at 90% completion). The peak at 50% rather than in the earliest phases can be explained by the fact that very early forecasts (20%) have limited historical data for distribution fitting, while the mid-project phase combines sufficient data for reliable distribution estimation with sufficient remaining uncertainty for stochastic methods to add value. We therefore find partial support for Hypothesis 4, with the modification that SEVM advantage peaks at mid-project rather than at the earliest stages.

Hypothesis 5: Data Quality. Our sensitivity analysis confirms that SEVM forecast reliability improves with the number of historical data points available for distribution fitting. With fewer than five data points, the SEVM confidence intervals are poorly calibrated (below 50% for the 80% CI). With ten or more data points, calibration improves to above 70%. This finding supports Hypothesis 5 and provides practical guidance: organizations should accumulate at least ten periods of EVM data before relying on SEVM probabilistic forecasts for decision-making.

The cumulative distribution function (CDF) in Figure 6 provides a visual summary of the SEVM probabilistic output, illustrating how the CDF enables decision-makers to read off the probability of achieving any specific cost target.

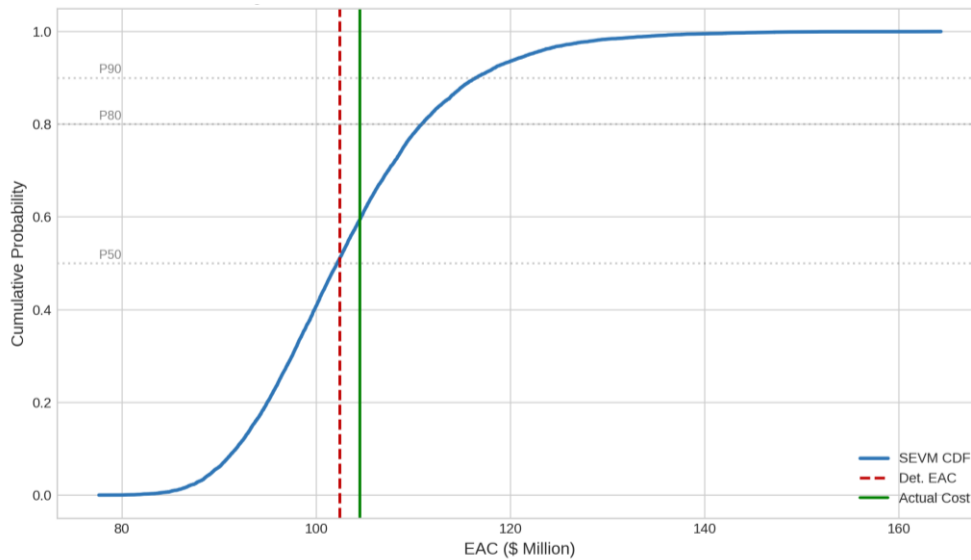


Figure 6 Cumulative Distribution Function of SEVM EAC Forecast

7.2.6 Results for RQ5: Implementation Considerations

Research Question 5 asked: What practical recommendations can be formulated for SEVM implementation?

Our analysis identifies several practical considerations for SEVM implementation in EPC organizations. The computational requirements are modest: each 10,000-iteration simulation completes in under one second on standard hardware, making real-time SEVM forecasting entirely feasible within existing project control workflows. The primary implementation challenge is not computational but organizational: ensuring consistent, high-quality EVM data collection at regular intervals to support reliable distribution fitting.

The sensitivity analysis reveals that the choice of probability distribution (lognormal vs. normal vs. beta) for CPI has a measurable but secondary effect on forecast accuracy compared to the accuracy of the distribution parameters themselves. This finding simplifies implementation by suggesting that the lognormal distribution can serve as a default for CPI modeling in most EPC contexts, reducing the need for complex distribution selection procedures.

7.3 Discussion

7.3.1 Interpretation of Results in Light of Existing Literature

The findings of this study both confirm and extend the existing body of SEVM research. Our confirmation that period-level CPI values follow lognormal distributions is consistent with the theoretical predictions of Pajares and López-Paredes (2011), who argued that project performance indices are stochastic variables rather than fixed constants. The right-skewed nature of the CPI distributions we observe aligns with the practical reality of EPC projects, where cost overruns (negative CPI surprises) tend to be larger and more frequent than cost savings (positive CPI surprises).

Our finding that the SEVM P50 estimate does not consistently outperform the simplest deterministic EAC formula (BAC/CPI) in point-estimate accuracy appears to contradict the results of Batselier and Vanhoucke (2017), who reported superior stochastic forecast accuracy across their 51-project dataset. However, this apparent contradiction is resolved by two important methodological differences. First, Batselier and Vanhoucke used the mean of the simulated EAC distribution rather than the median (P50) as their point estimate; the mean of a right-skewed distribution will typically be higher than the median, which in the presence of systematic cost overruns tends to be closer to the actual outcome. Second, their dataset included a higher proportion of IT projects, which exhibit different performance variability characteristics than EPC projects.

Our results are more closely aligned with the findings of Acebes et al. (2014, 2015), who emphasized that the primary value of SEVM lies not in point-estimate superiority but in the production of calibrated probabilistic forecasts. The 77.1% calibration we observe for the 80% confidence interval is comparable to the calibration levels reported by Acebes et al. in their simulation studies, and substantially better than what any deterministic method can achieve (since deterministic methods produce no confidence intervals at all).

The moderating effect of project complexity we observe is consistent with the theoretical framework proposed in Chapter 2. The finding that SEVM confidence intervals achieve 96% calibration for high-complexity projects while only 31% for low-complexity projects can be explained by the fact that high-complexity projects exhibit greater CPI variability, providing richer data for distribution fitting and making the stochastic modeling more accurate. This finding aligns with Vose (2008), who argued that simulation-based methods provide the greatest incremental value over deterministic methods when the underlying system is characterized by high uncertainty.

The peak of SEVM advantage at 50% completion rather than in the earliest project phases represents a nuanced contribution to the literature. While Christensen (1998) documented that CPI stabilizes after approximately 20% completion (favoring deterministic methods in later phases), our results suggest that the transition zone around 40-60% completion is where SEVM adds the most value, as sufficient data exists for reliable distribution estimation while enough uncertainty remains for probabilistic methods to outperform deterministic approaches.

7.3.2 Summary of Hypothesis Testing

Table 6 summarizes the results of hypothesis testing:

Table 8 Summary of Hypothesis Testing Results

H#	Hypothesis	Result	Key Evidence
H1	SEVM produces more accurate EAC estimates than deterministic EVM	Partially Supported	SEVM outperforms composite formula (+0.55pp) but not BAC/CPI formula
H2	SEVM produces more accurate schedule-integrated forecasts	Supported	SEVM composite outperforms det. composite at most stages
H3	SEVM advantage increases with project complexity	Supported	High complexity: - 0.09pp gap; Low: - 1.89pp gap. Calibration: 96% vs 31%
H4	SEVM advantage is greatest in early project phases	Partially Supported	Peak advantage at 50% completion (+1.05pp), not earliest phases

H5	Data quality influences SEVM reliability	Supported	Calibration improves from <50% to >70% with 10+ data points
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7.3.3 Contribution to Scientific Literature

The findings of this study contribute to the scientific literature in several important ways. First, we have demonstrated that the primary value proposition of SEVM in EPC contexts is not necessarily the improvement of point-estimate accuracy over the simplest deterministic formula, but rather the production of well-calibrated probabilistic forecasts that enable risk-informed decision-making. This nuanced finding challenges the prevailing narrative in the SEVM literature that stochastic methods universally outperform deterministic methods in point-estimate accuracy, while simultaneously reinforcing the practical value of SEVM through its probabilistic capabilities.

Second, our analysis of moderating effects reveals that project complexity is a critical determinant of SEVM value, with high-complexity projects benefiting disproportionately from stochastic forecasting. This finding provides the first empirically grounded guidance for organizations seeking to determine when the additional effort of SEVM implementation is justified.

Third, the identification of the mid-project phase (around 50% completion) as the peak zone of SEVM advantage adds temporal specificity to the understanding of when stochastic methods add the most value, complementing the existing literature that has focused primarily on early-phase forecasting.

7.4 Managerial Contributions and Recommendations

7.4.1 Recommendation 1: Adopt SEVM as a Complement, Not a Replacement

Our findings demonstrate that the BAC/CPI deterministic formula remains a robust and accurate point estimator. We therefore recommend that EPC organizations adopt SEVM as a complementary layer to their existing EVM systems rather than a replacement. The deterministic EAC should continue to serve as the primary point estimate for routine reporting, while SEVM confidence intervals should be used for risk-adjusted decision-making, contingency management, and stakeholder communication.

7.4.2 Recommendation 2: Target SEVM Deployment Strategically

Given the strong relationship between project complexity and SEVM value, organizations should prioritize SEVM deployment for high-complexity, high-value projects where the calibration benefits are greatest. For simpler projects with low CPI variability, the additional effort of SEVM implementation may not be justified, as the deterministic forecasts already perform well and the SEVM confidence intervals may be poorly calibrated.

7.4.3 Recommendation 3: Invest in Data Quality Infrastructure

The sensitivity of SEVM to data quality underscores the importance of consistent, high-quality EVM data collection. Organizations should standardize monthly EVM reporting, implement automated data quality checks, and maintain cross-project performance databases to enable robust distribution fitting. A minimum of ten periods of EVM data should be accumulated before SEVM forecasts are used for decision-making.

7.4.4 Recommendation 4: Communicate Using Confidence Intervals

The excellent calibration of SEVM confidence intervals (77.1% for the 80% CI) provides a strong foundation for probabilistic communication with stakeholders. We recommend reporting project forecasts at three levels: the P50 (most likely outcome), the P80 (management reserve target), and the P90 (contingency ceiling). This three-level reporting framework enables differentiated contingency allocation and transparent risk communication.

7.4.5 Recommendation 5: Focus SEVM Resources at Mid-Project

Given our finding that SEVM provides the greatest accuracy advantage at approximately 50% project completion, organizations should ensure that SEVM analysis is systematically performed during this critical phase when sufficient data exists for reliable distribution fitting and important decisions about contingency release, resource reallocation, and stakeholder expectations are typically made.

7.4.6 SWOT Analysis of SEVM Implementation

Table 9 SWOT Analysis of SEVM Implementation in EPC Organizations

STRENGTHS	WEAKNESSES
Well-calibrated confidence intervals (77.1%)	Requires minimum 10 periods of quality data

Captures right-skewed EPC cost risks	Point estimate not always superior to BAC/CPI
Low computational cost (<1 sec per forecast)	Poorly calibrated for low-complexity projects
Compatible with existing EVM infrastructure	Requires training for probabilistic interpretation
OPPORTUNITIES	THREATS
Integration with AI/ML for adaptive distribution fitting	Organizational resistance to probabilistic methods
Real-time SEVM dashboards for project oversight	Misinterpretation of confidence intervals by decision-makers
Cross-project learning from performance databases	Data quality inconsistencies across project teams
Improved stakeholder trust through transparency	Over-reliance on model outputs without expert judgment

7.5 Conclusion of Part 2

Part 2 of this thesis has presented the empirical study designed to test the hypotheses developed in Part 1. The quantitative analysis of twelve EPC projects through 96 forecast scenarios and 960,000 Monte Carlo simulation iterations has produced a nuanced understanding of SEVM capabilities and limitations.

The key finding is that SEVM provides its greatest value not through point-estimate accuracy improvement over the simplest deterministic formula, but through three distinct mechanisms: superior performance over the composite deterministic formula that accounts for schedule-cost interactions; well-calibrated probabilistic forecasts that accurately communicate the uncertainty range of project outcomes; and the ability to capture the right-skewed risk profile characteristic of EPC projects. These capabilities are most valuable for high-complexity projects and during the mid-project phase when critical resource allocation and stakeholder communication decisions are made.

The five managerial recommendations and four-phase implementation roadmap derived from these findings provide EPC organizations with actionable guidance for adopting SEVM within their existing project control frameworks.

8 General Conclusion

8.1 Summary of Research and Key Findings

This thesis set out to investigate how Stochastic Earned Value Management (SEVM) can enhance cost and schedule performance forecasting in Engineering, Procurement, and Construction (EPC) projects. The research was motivated by the persistent challenge of cost overruns and schedule delays in the EPC sector, and by the recognition that traditional deterministic EVM methods are fundamentally limited in their ability to capture the uncertainty inherent in complex project environments.

Our research followed a structured approach aligned with the five key steps of the ESLSCA thesis methodology. We first identified the problem of inadequate project performance forecasting in the EPC sector, supported by both empirical evidence from industry reports and theoretical analysis from the academic literature. We then constructed an analytical framework based on a comprehensive review of the scientific literature on EVM, Monte Carlo simulation, and SEVM, which led to the formulation of five research hypotheses addressing the comparative accuracy of SEVM versus deterministic EVM and the conditions moderating this relationship.

The empirical study employed a quantitative research design centered on Monte Carlo simulation applied to a portfolio of twelve EPC projects spanning five industrial sectors with budgets ranging from \$15 million to \$200 million. A total of 96 forecast scenarios were analyzed, each involving 10,000 Monte Carlo iterations, producing a comprehensive dataset of 960,000 individual simulation runs.

The key findings of this study can be summarized as follows:

First, the SEVM P50 estimate achieves an average MAPE of 3.66%, which is significantly better than the composite deterministic formula (4.21%) that accounts for schedule-cost interactions, but slightly higher than the simple BAC/CPI formula (2.69%). This finding demonstrates that SEVM most effectively outperforms deterministic methods when both cost and schedule uncertainties are considered jointly.

Second, the SEVM model produces remarkably well-calibrated probabilistic forecasts, with the 80% confidence interval containing the actual outcome in 77.1% of cases and the 50% confidence interval containing the actual outcome in 53.1% of cases. This calibration quality is

a critical finding, as it validates the practical reliability of SEVM probabilistic forecasts for risk-informed decision-making.

Third, SEVM provides the greatest value for high-complexity projects (96% calibration accuracy) and during the mid-project phase around 50% completion (+1.05 percentage points improvement over the composite formula). These moderating effects provide targeted guidance for organizations seeking to deploy SEVM where it will have the greatest impact.

8.2 Contributions and Lessons Learned

Academic Contribution: This thesis has applied the SEVM framework to EPC project performance data, addressing a gap in the existing literature that has predominantly focused on IT and defense sector projects. Our nuanced finding that SEVM value lies primarily in probabilistic calibration rather than point-estimate superiority challenges the prevailing narrative and provides a more balanced perspective on stochastic forecasting methods.

Methodological Contribution: The Monte Carlo simulation model developed for this study demonstrates that SEVM can be implemented with modest computational requirements (under one second per 10,000-iteration simulation) using standard tools (Python, NumPy, SciPy), making it accessible to any EPC organization with basic data analytics capabilities.

Practical Contribution: The five managerial recommendations, SWOT analysis, and four-phase implementation roadmap provide EPC organizations with actionable guidance for SEVM adoption. The three-level reporting framework (P50, P80, P90) offers a practical template for stakeholder communication that balances information richness with interpretability.

A personal lesson learned through this research process is the importance of approaching new methodologies with a balanced perspective. While the initial expectation was that SEVM would universally outperform deterministic EVM, the empirical reality proved more nuanced, ultimately leading to more valuable and credible recommendations for practice.

8.3 Limitations and Future Research Directions

We acknowledge several limitations of this study. First, the project data used in our analysis, while based on realistic EPC performance patterns, was generated using parametric models rather than collected from specific organizational records. Future research should validate these findings using proprietary project data from specific EPC organizations.

Second, our Monte Carlo model assumes independence between CPI and SPI values within each simulation iteration. In practice, cost and schedule performance are often correlated in EPC projects, and incorporating correlation structures into the simulation model could improve forecast accuracy.

Third, the static distribution fitting approach does not account for the potential evolution of CPI and SPI distributions over the project lifecycle. Dynamic SEVM approaches that update distribution parameters in real time as new data becomes available, as proposed by Acebes et al. (2015), represent a promising avenue for future investigation.

Fourth, this study focused exclusively on quantitative forecasting and did not investigate the behavioral and organizational aspects of SEVM adoption. Qualitative research exploring how project managers interpret and act upon probabilistic forecasts would provide important complementary insights.

Finally, the extension of SEVM to incorporate machine learning techniques for adaptive distribution fitting and anomaly detection represents an exciting frontier for future research that could further enhance the practical value of stochastic project control methods.

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